Income Heterogeneity of Immigrants and House Price Dynamics

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Abstract

This paper examines how the income composition of immigrant inflows affects local house price growth in the Netherlands. Using administrative microdata from 2006 to 2020, we distinguish immigrants into four income groups and estimate their effects on property values and transaction-based house prices. To address potential endogeneity in residential location choices, we employ a shift-share instrumental variable strategy based on historical settlement patterns interacted with national immigration shocks by origin and income group. We find heterogeneity depending on immigrant income, geographic location, and the time horizon analyzed. In the short run, middle-low-income immigrant inflows increase annual property-value growth by approximately 0.17–0.20 percentage points, while low-income inflows reduce it by approximately 0.05–0.06 percentage points. These patterns appear primarily in less urban municipalities. Medium-run estimates suggest weaker subsequent property-value growth, consistent with residential adjustment. Native population responses vary across immigrant income groups, but the evidence does not establish native out-migration as the primary mechanism behind the estimated house price effects. Overall, the findings show that aggregate estimates can mask important differences in the local housing-price effects of immigration.

Keywords: Housing prices, immigration, urban areas
JEL Codes: J61, R23, R31

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Results in this paper are based on non-public microdata from Statistics Netherlands (CBS). In principle, part of these microdata is accessible for statistical and scientific research. For further information: microdata@cbs.nl.

1 Introduction

Immigration and housing affordability are closely linked in public and policy debates. As immigrants settle in a country, they contribute to local housing demand, potentially putting upward pressure on house prices, particularly in markets where housing supply responds slowly to changes in demand (Saiz, 2003, 2007). Yet the effect of immigration on house prices is not straightforward. While some empirical studies find that immigration increases housing costs (Gonzalez & Ortega, 2013; Degen & Fischer, 2017), others document negative or statistically insignificant effects (Sá, 2015; Accetturo et al., 2014). Understanding the relationship between immigration and house prices therefore remains important for assessing the local consequences of population inflows and their implications for housing affordability.

An important consideration in this relationship is that immigrants differ in their economic resources and housing needs. Immigrants with different income levels may enter different segments of the housing market, choose different neighborhoods, and face different constraints in accessing homeownership. Higher income arrivals may increase demand for relatively expensive properties and influence the attractiveness of residential locations to other households. Lower-income arrivals may concentrate in more affordable neighborhoods, where the effects of additional demand depend on the availability and quality of the housing stock, local amenities, and the responses of incumbent residents. Moreover, changes in neighborhood composition can affect property values beyond immigrants' direct demand for housing. Residential sorting may alter the composition of local demand, while the relocation of incumbent households can offset some of the initial pressure generated by population inflows. Consequently, the effect of immigration on house prices need not be uniform across income groups, neighborhoods, or time horizons.

This paper examines how the income composition of immigrant inflows affects local house price growth in the Netherlands. We classify arriving immigrants into four income groups based on the destination municipality's income distribution: low, middle-low, middle-high, and high income. We then estimate the price impact of changes in the local population share for each group, as well as for all immigrants combined. Our analysis focuses on two complementary measures of residential property values: the tax assessment property value (*Waardering Onroerende Zaken* (WOZ))-the annual administrative valuation used for property taxation-, and a repeat-sales price index constructed from housing transactions. This combination allows us to examine changes in assessed property values across the housing stock alongside changes in realized transaction prices for properties sold more than once. We further investigate whether estimated effects differ between more urban and less urban municipalities and whether medium-term price dynamics are consistent with residential adjustment among incumbent residents.

The Netherlands provides an ideal empirical setting to analyze these dynamics. Between 2006 and 2020, average Dutch selling prices appreciated by 31%, alongside sustained foreign immigration. The Dutch experience reflects broader trends across high-income economies, where rapid housing price appreciation over the same period coincided with substantial immigration inflows and heightened housing affordability pressures.[1] Moreover, the Dutch administrative data infrastructure allows us to observe these changes at a localized level. By merging individual population registers from Statistics Netherlands (CBS) with administrative real estate transaction records from the Land Registry (*Kadaster*), we follow 2,349 four-digit postcodes across 341 municipalities—covering over 80% of the national population and housing stock. These data allow us to link individual demographic and income records directly to neighborhood-level property values, providing a clear view of how the income composition of immigrant inflows shapes local housing prices.

Identifying the causal effect of immigrant inflows on local house prices is challenging because residential location choices are endogenous. Immigrants may be attracted to neighborhoods with improving economic prospects, better amenities, or rapidly appreciating property values. Conversely, immigrants with fewer economic resources may disproportionately settle in areas with relatively low housing costs or weaker price growth. These patterns create a potential correlation between immigrant inflows and local housing-market developments even in the absence of a causal effect of immigration. To address this concern, we employ a shift-share instrumental variable strategy. The intuition is to predict which postcodes are more exposed to subsequent immigration inflows using settlement patterns established before our house price estimation period. Specifically, we predict annual postcode-level immigrant inflows for each of the five income tiers by combining two components: each postcode's 1995 share of the national population from a given origin region, which measures its baseline connection to that origin group, and the subsequent change in national immigration from that origin region, measured separately by immigrant income category. By combining these two terms, the instrument predicts larger immigration changes in postcodes that historically had stronger settlement links to origin regions experiencing larger subsequent national inflows. This strategy relies on established immigrant network mechanisms. Newly arriving immigrants settle near existing co-national communities, where ethnic networks provide information on housing and job opportunities, facilitate access to origin-specific goods and services, and lower language barriers (Munshi, 2003; Zorlu & Mulder, 2008). By constructing separate instruments for immigrant income groups, we

[1]Over this period, housing prices grew by 73% in Germany, 43% in Portugal, 23% in the United States, and 19% in Denmark, while foreign population shares expanded significantly across Western Europe (e.g., net foreign immigration grew by 69% in Germany and legal foreign residents by 99% in Portugal). European housing data are from Eurostat's House Price Index (Eurostat), German immigration statistics from Destatis (Destatis), Portuguese foreign resident data from PORDATA (PORDATA), and US price indices from FHFA (FRED).

obtain predicted variation in local inflows that reflects both historical settlement patterns and national migration shocks.

To evaluate how these predicted shocks impact local housing markets, we estimate a panel fixed-effects model at the postcode-year level. Our baseline specification regresses annual changes of neighborhood house prices (in natural logarithm) on predicted immigrant inflows across the five income tiers, absorbing unobserved time-invariant spatial heterogeneity via postcode fixed effects and common macroeconomic shocks via year fixed effects. To ensure that our estimates isolate housing demand driven by immigration rather than broader local trends, we control for lagged changes in local labor market conditions, working population growth, and housing supply, while allowing neighborhoods with different initial economic characteristics to follow flexible time trends. The core identifying assumption is that, conditional on these controls, a postcode's historical settlement share interacted with national origin-by-income shocks affects local house price growth solely through its impact on subsequent immigrant inflows. The combination of pre-period settlement shares, administrative microdata, and granular spatial fixed effects provides a credible framework for isolating the causal impact of income-differentiated immigration.

Our results show that treating immigrants as a homogeneous population masks heterogeneity in local house price responses. In the baseline instrumental variable estimates using the tax assessment property value, inflows of middle-low-income immigrants are associated with higher property value growth, whereas low-income immigrant inflows have a negative effect on local valuations. Evaluated at the mean change in the relevant immigrant population share, the estimated middle-low-income effect corresponds to an increase of approximately 0.17–0.20 percentage points in annual property value growth. The corresponding effect for low-income immigrants is a reduction of approximately 0.05–0.06 percentage points. By contrast, the estimates for the aggregate immigrant population and for the upper-income groups are less precisely estimated in the full sample, preventing firm conclusions about their average effects. The findings therefore point not to a uniform positive or negative effect of immigration, but to substantial heterogeneity across the income distribution. Results based on the repeat-sales index broadly corroborate the positive association for middle-low-income inflows, although estimates for other groups are more sensitive to specification, highlighting the importance of distinguishing administrative valuations from transaction-based measures of house price dynamics.

The effects also vary geographically. The positive tax assessment value responses are concentrated in less urban municipalities and in areas outside the four largest Dutch cities. In these locations, the estimates indicate positive effects for both high-income and middle-low-income immigrant inflows. In contrast, the corresponding estimates for the largest urban housing markets are generally statistically indistinguishable from zero. These differences suggest that the

relationship between immigration and house prices depends not only on the characteristics of incoming residents but also on the local market in which additional demand occurs. One possible interpretation is that demographic changes have different marginal effects across housing markets with different initial immigrant shares, housing-stock compositions, and price dynamics. However, the estimated geographic heterogeneity does not, by itself, identify which of these mechanisms accounts for the differences.

Finally, we examine house price dynamics over a longer horizon and investigate whether changes in the native population provide evidence of residential adjustment following immigrant inflows. The medium-term estimates indicate weaker subsequent property value growth in neighborhoods experiencing an increase in the immigrant share, a pattern consistent with the possibility that initial demographic shocks are followed by changes in residential sorting. Native out-migration is one potential channel through which the composition of local housing demand may adjust over time. These results complement the short-run estimates by highlighting that the relationship between immigration and house prices may evolve as households respond to changes in neighborhood composition. Nevertheless, the evidence should be interpreted cautiously: a negative medium-term price response and changes in the native population are consistent with residential adjustment, but do not establish that native out-migration is the sole mechanism driving subsequent price dynamics.

This paper contributes to three strands of the literature. First, it contributes to research on immigration and housing markets by examining heterogeneity in immigrant inflows according to income, rather than treating immigration as a homogeneous population shock (Akbari & Aydede, 2012; Gonzalez & Ortega, 2013; Sá, 2015; Braakmann, 2019). By separating incoming immigrants into income groups, we show that aggregate estimates can mask opposing responses within the same national housing market. Second, the paper contributes to the literature on the spatial incidence of immigration by using four-digit postcode data to identify localized price responses and examine heterogeneity across urban and less urban settings (Accetturo et al., 2014; Mussa, Nwaogu, & Pozo, 2017). Third, we contribute to research on residential sorting and the longer-run adjustment of housing markets by examining price dynamics alongside changes in the native population. In doing so, we connect the immediate effects of immigrant inflows to the possibility that household relocation and changes in neighborhood composition shape subsequent local housing market outcomes.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data and presents descriptive statistics. Section 4 introduces the empirical framework and shift-share identification strategy. Section 5 presents the main results, including robustness checks and heterogeneity across local housing markets. Section 6 discusses the findings and concludes.

2 Related literature

This paper contributes to the literature on immigration and housing markets by examining how the income composition of immigrant inflows shapes local house price dynamics. Immigration acts as a positive population shock that increases local housing demand and exerts upward pressure on prices and rents, particularly when supply is inelastic (Saiz, 2003, 2007). Most empirical studies find that immigrant inflows boost housing prices—such as Akbari and Aydede (2012) in Canada, Gonzalez and Ortega (2013) in Spain, Degen and Fischer (2017) in Switzerland, Hyslop et al. (2019) in New Zealand, and Moallemi and Melser (2020) in Australia. However, studies in other settings document negative or negligible price responses in target neighborhoods, such as Sá (2015) and Braakmann (2019) in the UK, as well as Accetturo et al. (2014) in Italy. Reconciling these contrasting empirical findings requires evaluating three key dimensions: immigrant heterogeneity, spatial and temporal scales of adjustment, and native behavioral sorting.

Most of the literature treats immigrants as a homogeneous group, despite differences in purchasing power, skills, and housing preferences that may shape the local housing markets they enter (Albert & Monras, 2022). Sá (2015) finds that negative house price effects in the United Kingdom are concentrated in areas receiving less-educated immigrants, while Cabral and Steingress (2026) finds larger shelter price effects in US counties receiving more highly educated immigrants, particularly where building permits are scarce. Related work also highlights the importance of immigrant characteristics in shaping local economic outcomes (Monras, 2020). This paper extends the literature by classifying incoming immigrants into income quartiles at the municipality level using Dutch administrative data.

The geographic scale of analysis can explain differences in estimated house price effects. Aggregate estimates capture changes in overall housing demand, whereas neighborhood-level estimates also reflect the spatial redistribution of households. In Italy, Accetturo et al. (2014) find that immigration raises city-level house prices but reduces relative price growth in districts receiving larger inflows. Mussa, Nwaogu, and Pozo (2017) identify price effects in both immigrant-receiving and neighboring US metropolitan areas, highlighting the potential for spatial spillovers. These patterns may also depend on local housing supply constraints, which determine how strongly additional demand translates into prices rather than construction (Glaeser, Gyourko, & Saiz, 2008; Saiz, 2010). Using four-digit postcode data in the Netherlands, this paper estimates localized price responses.

Furthermore, housing market adjustments unfold over distinct temporal horizons. In the short run, supply constraints and construction lags dictate that demand shocks translate directly into price increases (Saiz, 2003; Unal, Hayo, & Erol, 2024). Over longer horizons, market clearing occurs through filtering, residential construction, endogenous labor supply expansion (Monras,

2020), and shifts in local amenities (Sanchis-Guarner, 2023). As emphasized by Sanchis-Guarner (2023), capturing the full equilibrium effect of immigration requires evaluating medium-term spatial adjustments rather than immediate price reactions. Following this insight, we evaluate house price growth over a three-year horizon alongside annual changes, isolating structural medium-term adjustments from short-term market noise.

Finally, the mechanism proposed to explain negative localized price responses is native out-migration or residential displacement (commonly termed "native flight") (Sá, 2015; Fernández-Huertas Moraga, Ferrer-i-Carbonell, & Saiz, 2019; Boje-Kovacs et al., 2024). Inflows can induce pre-existing residents to relocate or avoid target neighborhoods due to changing amenity preferences or perceived social distance, generating a secondary supply outflow (Hatton & Tani, 2005; Larkin et al., 2019; Andersson, Malmberg, & Vitali, 2021). However, the empirical magnitude of native displacement remains actively debated, with recent neighborhood-level evidence suggesting incumbent out-migration responses can be minimal (Peruffo & Check, 2026). Native sorting also depends on skill and socio-economic complementarities (Card & DiNardo, 2000; Mocetti & Porello, 2010). We contribute to this mechanism literature by tracking changes in the native population share at the postcode level across incoming immigrant groups, testing whether affluent foreign inflows attenuate native flight relative to lower-income arrivals.

3 Data

To examine how neighborhood house prices respond to immigrant inflows, we combine individual-level administrative registers from Statistics Netherlands (CBS) with real estate transaction records from the Land Registry (Kadaster) for the years 1995 to 2022. While our demographic and overall housing market panel spans 1995 to 2022, granular tax register data on individual incomes are consistently available starting in 2006. Consequently, our analysis of immigrant heterogeneity by income group focuses on the 2006–2020 period.

Spatial Unit and Sample Selection. We conduct our analysis at the four-digit postcode level (PC4), which we use interchangeably with "neighborhood" and "postcode". With an average territory of 2.5 km^2, four-digit postcodes represent a localized geographic scale where residents directly experience local demographic changes and neighborhood amenities (Veldhuizen, Stronks, & Kunst, 2013). To link our demographic data with rich neighborhood characteristics—such as population density, local infrastructure, and housing stock composition—we merge our panel with the regional CBS database (*Kerncijfers Wijken en Buurten* (KWB)).

To construct a reliable panel, we apply a series of sample selection criteria, summarized in Table 1. First, we restrict our sample to a balanced panel of four-digit postcodes that exist continuously between 1995 and 2022. Second, we drop postcodes with zero inhabitants in any given

year. Third, to avoid outlier distortions, we exclude postcodes experiencing year-on-year immigration shifts exceeding 100%. Finally, we impose a minimum population threshold in 1995 at the 30th percentile of the national distribution (1,011 inhabitants) to filter out sparsely populated industrial or rural tracts.

Table 1: Number of postcodes by selection step

Filtering Step	N postcodes
All postcodes	4,056
> 0 inhabitants	3,978
Immigrant growth rate < 100%	3,416
30th pct population (1995)	2,349

Note: Each line represents a successive sample selection. Postcode refers to a four-digit postcode.
Source: CBS Microdata

Our final estimation sample comprises 2,349 postcodes, representing an average annual stock of roughly 4,904 inhabitants older than 18 years old (including 514 immigrants) and 4,133 dwellings per postcode. This sample covers over 80% of the total Dutch population, immigrant population, and national housing stock.

Immigration. Our primary regressor of interest is the change in the local immigrant population as a share of the total postcode population in the previous year. Drawing from CBS administrative registers, we define immigrants as foreign-born individuals who settle in a Dutch municipality from abroad at age 18 or older. Our empirical analysis focuses on immigrants moving to the Netherlands for labor, family, and student motives.[2] To account for heterogeneity in purchasing power, we classify incoming immigrants into five income groups: total immigrant population, low, middle-low, middle-high, and high income. We measure income using the total wage on which wage tax and social insurance contributions are calculated obtained from the Tax and Customs Administration (*Belastingdienst*), which covers the universe of registered residents in the Netherlands and is available continuously from 2006 onwards. Income groups are defined based on an immigrant's observed annual income in the year of arrival, relative to the income distribution of the total population in the destination municipality. Each immigrant's income group is assigned at arrival and held fixed thereafter, so that subsequent changes in an individual's income do not mechanically alter their classification. Individuals with zero observed income or missing income are assigned to the lowest income category. Income at arrival provides an observable measure of immigrants' relative economic resources, but it does not fully capture their purchasing power in the housing market. In the Netherlands, access to

[2]Asylum seekers and refugees residing in central reception facilities are excluded because their spatial placement is governed by state quota allocations rather than voluntary market choices. Dröes and Koster (2023) consider granular corridors around refugee settlement centers in the Netherlands and find locally concentrated decrease in house prices in areas close to refugee pathways.

mortgage credit depends not only on the value of the property relative to the loan but also on income-related borrowing constraints and households' broader financial circumstances. These income measures capture relative income position at arrival, but they do not capture wealth, savings, household size, existing debt, or differences in access to credit.

Housing Prices. To measure local housing market responses, we employ two complementary price indicators: the municipal property tax assessment value (*Waardering Onroerende Zaken*, WOZ) and a repeat-sales price index. The WOZ value, assessed annually by municipalities for tax purposes, estimates the market value of every residential property using structural features, square footage, and recent transactions of comparable units within the neighborhood (see Appendix B.3 for detailed information on the WOZ calculation methodology).

Second, we construct a Case and Shiller (1987) repeat-sales price index using the universe of real estate transaction prices for existing owner-occupied dwellings recorded by the Land Registry, setting 2015 as the base year. The repeat-sales methodology provides the key advantage of controlling for unobserved, time-invariant housing quality by holding property-specific characteristics constant across transactions.[3] This index tracks realized market prices across 56,500 to 126,500 annual transactions (averaging 40–50 sales per postcode-year).

Comparing these two indicators offers distinct empirical advantages. The WOZ value provides full-population coverage across all residential properties annually, though with an administrative lag reflecting past market trends. Because municipal tax valuations (WOZ) reflect the aggregate average value of the housing stock—capturing structural upgrades, expansions, and high-end residential development—they absorb both price shifts and quality improvements. In contrast, the repeat-sales index captures immediate, realized transaction prices—specifically overrepresenting more frequently traded housing types.

Our analysis focuses on owner-occupied housing prices rather than rents. Although our WOZ-based outcome captures assessed property values for properties covered by the data, including rental properties where present, it does not measure rents paid by tenants. We do not directly estimate the effects of immigration on rental prices due to data availability constraints. Moreover, the unregulated private rental sector accounts for approximately 10–13% of the Dutch housing stock, while owner-occupied housing represents a substantially larger segment of the market. We therefore focus on owner-occupied property values, for which we can construct consistent measures at the required spatial resolution. This choice allows us to study an important segment of the housing market, but it does not capture potential effects of immigration on rents or on households in the rental sector.

[3]Hedonic regression models represent a common alternative for constructing house price indices. However, administrative Land Registry transaction data lack the granular property-level structural attributes required for a hedonic specification.

3.1 Descriptive Statistics

Our final estimation sample comprises 2,349 four-digit postcodes across 341 Dutch municipalities, covering approximately 10.1 million adult residents and 9.7 million housing units at baseline (1995). Table 2 presents summary statistics across postcodes and years. The median neighborhood in our sample contains 5,450 adult inhabitants (older than 18) and a housing stock of 4,295 dwellings.

Table 2: Summary Statistics across Postcodes (2006–2020)

	Mean	St. Dev.	25th Pct.	Median	75th Pct.	N
Avg. WOZ value by PC4 (x 1,000)	239.2	89.5	182.0	226.4	276.9	32,487
Δ Avg. WOZ by PC4 (%)	1.7	5.7	-2.4	1.4	5.4	32,345
Repeat-Sales Index	109.6	20.9	100.0	107.2	116.9	32,895
Δ Repeat-Sales Index (%)	1.1	12.7	-4.3	1.5	6.8	32,846
Housing Stock	2,772	1,737	1,360	2,521	3,846	33,085
Δ Housing Stock (%)	22.7	342.4	-1	9	32	33,085
Share of rental units (%)	39.4	16.4	27.4	35.9	48.8	32,160
Share of owner-occupied units (%)	58.8	16.8	49.1	62.6	71.0	32,167
Population:						
Population age ≥ 18	4,914	2,846	2,544	4,510	6,796	33,085
Δ Population (%)	-4.7	863	-13	12	52	33,084
Immigrants age > 18	519	730.7	98	276	618	33,085
Δ Migrants (%)	-3	185	-1	5	18	33,084
Migrant share (%)	8.9	9.0	3.3	5.8	10.9	33,085
Avg. household size	2.2	0.3	2.1	2.3	2.4	32,649
Income:						
Avg. income per worker (x 1,000)	30.0	6.1	27.2	29.9	33.1	27,700
Share of low-income immigrants (%)	56.2	10.2	49.7	56.0	62.4	33,085
Share of middle-low-income immigrants (%)	15.9	5.9	12.0	15.6	19.2	33,085
Share of middle-high-income immigrants (%)	19.1	6.6	14.9	18.3	22.3	33,085
Share of high-income immigrants (%)	12.3	6.0	8.1	11.0	15.2	33,085

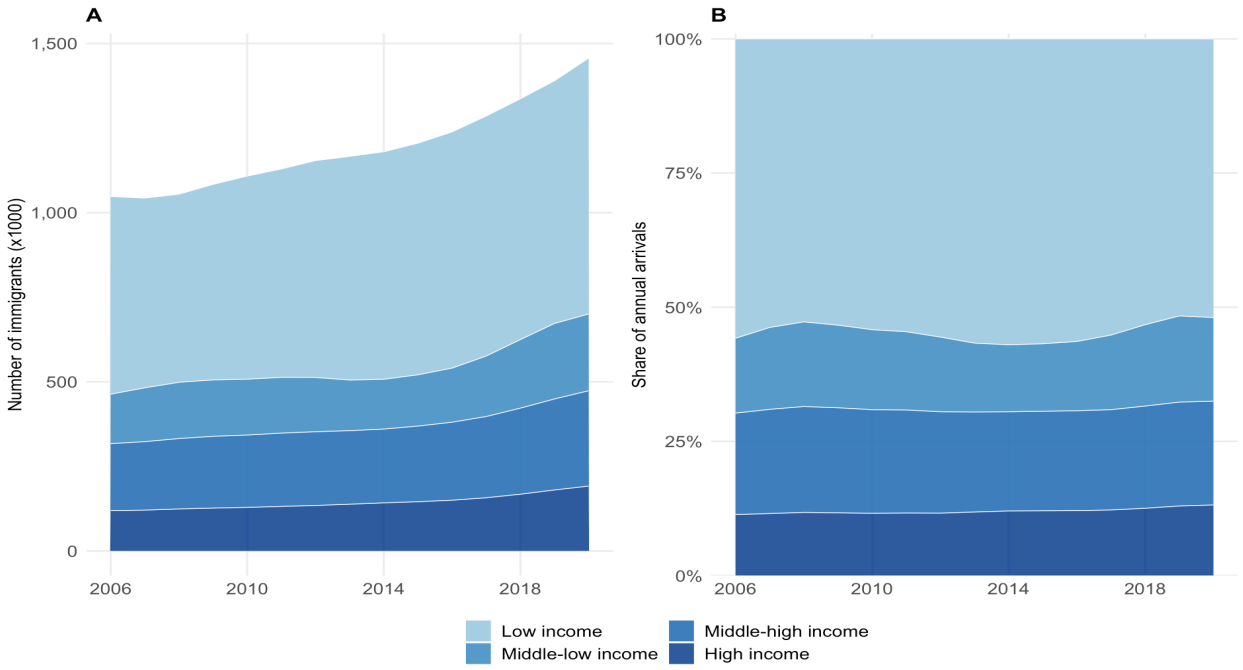
Notes: This table presents summary statistics across 2,349 four-digit postcodes and years. Δ denotes the year-on-year percentage change. Rental and owner-occupied shares are calculated relative to total local housing stock. Immigrant, low-income, and high-income shares are measured relative to the adult population as of January 1st. Low-income (high-income) population shares correspond to households in the first income quartile (fourth income quartile) of the municipality-level income distribution.
Sources: CBS Microdata; *Kerncijfers Wijken en Buurten* (KWB).

Immigrant shares exhibit substantial spatial and temporal variation across postcodes. On average, immigrants constitute 14% of a neighborhood's adult population, with an interquartile range spanning from 9% at the 25th percentile to 21% at the 75th percentile. This local variation in immigrant concentration provides the primary identifying variation for our empirical framework.[4]

[4]Figure D.1 in Appendix D illustrates the distribution of immigrant shares across municipalities in the estima-

Figure 1 documents the evolution of immigration to the Netherlands over the sample period and the changing income composition of incoming immigrants. Panel A shows the annual number of new adult immigrants overall and by income group, while Panel B reports each group's share of total arrivals. These descriptive patterns establish the scale of immigration and the extent to which the composition of incoming cohorts varies over time. They also provide context for the empirical strategy, which exploits variation in immigrant inflows across both income groups and locations.

Figure 1: Inflows of immigrants by income group



Notes: The left panel, A, displays the number of immigrants per year by income group; the right panel, B, shows the share of each immigrant income group out of total immigrants by year.
Sources: CBS Microdata; *Kerncijfers Wijken en Buurten* (KWB).

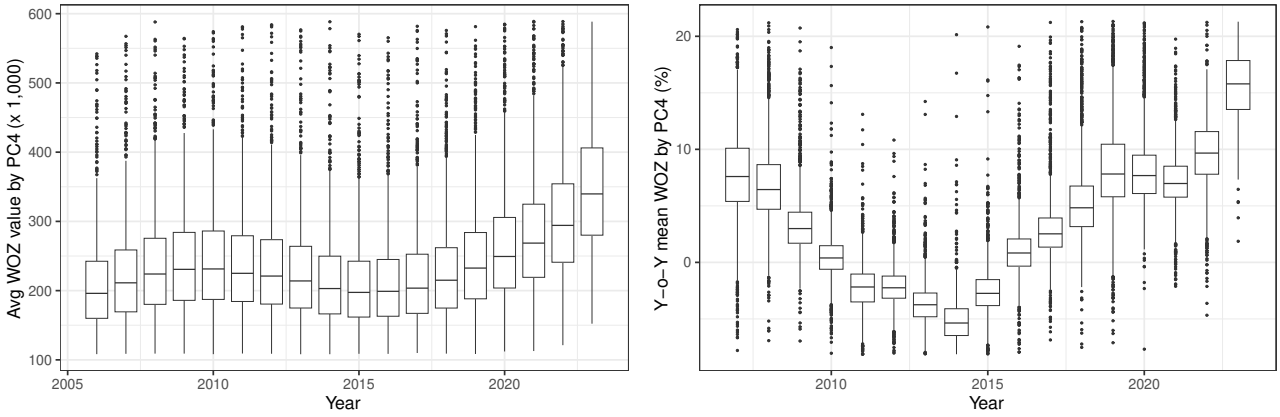
A potential concern with treating local immigrant arrivals as structural neighborhood demand shocks is the presence of short-term seasonal migration or hyper-mobile populations. However, administrative registry data indicate that immigration spells in our sample reflect durable residential decisions. Approximately 35% of foreign-born individuals who arrived between 1995 and 2015 remained continuously registered in the Netherlands through 2022 (Appendix Figure D.4). Conditional on completing an immigration spell, 75% of immigrants stay in the country for at least one year, with a median stay duration of approximately two years. Within the country, the median immigrant completes 2 to 3 housing spells, with 75% of individual address stays exceeding one year (Appendix Table D.5). The limited prevalence of short-term

tion sample in 1995.

seasonal stays confirms that immigrant arrivals represent sustained multi-year neighborhood demand shifts.

Figure 2 illustrates the level and annual growth rate of municipal property valuations (WOZ) across neighborhoods over time. Average WOZ values grew at an annual rate of 1.7% across postcodes, accompanied by pronounced macro-level fluctuations—most notably the market contraction between 2011 and 2015, followed by rapid appreciation.

Figure 2: Distribution of Municipal Property Values (WOZ) across Postcodes over Time



Notes: The left panel displays the distribution of average WOZ property valuations across postcodes; the right panel shows the year-on-year percentage growth rate. Box plots display the interquartile range (25th to 75th percentiles), with median lines and whiskers extending to 1.5 × IQR. Sample consists of 2,349 four-digit postcodes (PC4).
Sources: CBS Microdata; *Kerncijfers Wijken en Buurten* (KWB).

The repeat-sales transaction index exhibits a strong co-movement with the WOZ valuation series (Appendix Figure D.2), with a national year-on-year correlation coefficient of 0.86 (Appendix Figure D.3). The repeat-sales index exhibits slightly milder price contractions during the 2011–2015 downturn, consistent with transaction selection mechanisms where property owners delay sales during market troughs.

Finally, cross-sectional correlations highlight key structural determinants of local price appreciation. Over the sample period, WOZ property appreciation is positively correlated with the baseline share of high-income households ($\rho = 0.76$) and the share of owner-occupied housing ($\rho = 0.41$), while negatively correlated with the local rental housing share ($\rho = -0.42$). Furthermore, house price growth was significantly stronger in large municipalities ($> 100,000$ inhabitants; Appendix Table D.2), motivating our spatial heterogeneity analysis across urban and non-urban environments in Section 5.

4 Empirical Framework

We examine how the relationship between immigration and house prices varies with immigrants' position in the income distribution. Income shapes housing market dynamics through several channels. First, purchasing power influences tenure choice and access to credit, enabling higher income immigrants to enter sooner the owner-occupied market and exert upward pressure on house prices. Second, income influences neighborhood choice and amenity demand; as Duijn and Rouwendal (2021) establish, households across all demographics value a higher neighborhood share of high-income residents. High-income immigrants may thus generate positive neighborhood externalities—such as enhanced commercial amenities and local tax revenues—that are prized by native households. Third, income-driven sorting affects residential integration, as incumbent residents may prefer co-locating with higher income cohorts. Although individual preferences are unobserved, these mechanisms imply that high- and low-income immigrant inflows trigger distinct housing demand shocks and neighborhood spillovers. Motivated by these theoretical channels, we differentiate immigrant inflows by income tier and estimate their localized impact on house price growth.

Our baseline empirical specification is:

$$\Delta \log P_{i,t} = \beta^j \left(\frac{\Delta M^j_{i,t-1}}{\text{Pop}_{i,t-2}} \right) + \boldsymbol{\gamma}' X_{i,t-2} + \boldsymbol{\theta}'(W_i \times \delta_t) + \sigma_i + \delta_t + \varepsilon_{i,t}, \tag{1}$$

where the unit of observation is postcode i in year t. The dependent variable, $\Delta \log P_{i,t}$, denotes the annual change in the natural logarithm of house prices, evaluated separately using municipal tax valuations (WOZ) and our repeat-sales price index. The regressor of interest, $\Delta M^j_{i,t-1}/\text{Pop}_{i,t-2}$, measures the one-year lagged change in the stock of foreign-born immigrants in income category $j \in \{\text{Total}, \text{Low}, \text{Middle-Low}, \text{Middle-High}, \text{High}\}$, scaled by the baseline postcode population at time $t-2$. Immigrants are classified into municipal-level income quartiles in their year of arrival, with the low-income tier also incorporating non-working immigrants. The coefficient β^j captures the semi-elasticity of local house price growth with the scaled change in immigration inflows of income group j. To isolate the causal effect of immigrant inflows, we control for a vector of lagged, time-varying postcode characteristics $X_{i,t-2}$, comprising changes in the local unemployment rate, total housing stock, and working-age population (aged 15–64). Furthermore, postcodes with different baseline structures may experience differential long-run trends. We account for this by incorporating $W_i \times \delta_t$, interacting postcode attributes at the starting year of the sample, W_i—specifically the logarithm of average WOZ property value and logarithm of the median income—with year fixed effects. Because the specification is in first-differences, fixed postcode-level unobservables affecting price levels are eliminated. The

inclusion of postcode fixed effects, σ_i, further controls for postcode-specific trends in house price growth, ensuring that identification relies strictly on temporal deviations from a postcode's average price trajectory over our sample period. Year fixed effects, δ_t, absorb aggregate macro-financial shocks, such as national interest rate movements or housing policy changes. Finally, all regressions are weighted by 2006 postcode population to ensure results are representative of the overall housing market and to account for market-size heteroskedasticity. Standard errors are clustered at the municipality level to allow for spatial and serial correlation across postcodes.

4.1 Identification

The main identification challenge in estimating β^j via OLS is that immigrant location choices are endogenously determined. We highlight two main identification concerns. The first is reverse causality: local housing prices directly influence immigrants' location decisions. If lower income immigrants, constrained by housing affordability, systematically settle in postcodes with lower or stagnant property values, this induces a spurious negative correlation between immigrant inflows and house price growth. Conversely, if higher income immigrants select into areas experiencing rapid price appreciation—interpreting rising values as a signal of neighborhood quality—this creates a spurious positive correlation where appreciating neighborhoods attract more affluent migrants over time. Localized economic shifts, such as infrastructure investments, public transit expansions, or municipal amenity upgrades, simultaneously boost property valuations and attract new residents. Because these unobserved shocks ($\varepsilon_{i,t}$) are positively correlated with both house price growth and incoming foreign migration, standard OLS specifications suffer from an upward bias.

To estimate the causal effect of immigrant inflows across income groups on local house prices, we implement a shift-share instrumental variable (SSIV) strategy, following Bartik (1991) and Card (2001) and the more recent shift-share literature (Goldsmith-Pinkham, Sorkin, & Swift, 2020; Borusyak, Hull, & Jaravel, 2025). Specifically, we use the SSIV to predict annual postcode-level immigrant inflows for each of the five income tiers.

The intuition is to predict which postcodes are more exposed to subsequent immigration inflows using settlement patterns established before our house price estimation period.

The instrument combines two components. The first is each postcode's share of the national population from a given origin region in 1995. This historical share measures the postcode's pre-existing connection to that origin group. The second is the subsequent change in national immigration from that origin region, measured separately by immigrant income category. This national change captures the timing and scale of the inflow from each origin. By combining the two components, the instrument predicts larger immigration changes in postcodes that historically had stronger settlement links to origin regions experiencing larger subsequent national

inflows.

The approach is motivated by immigrant network effects: new arrivals often settle near established co-national communities, where they may obtain information about housing, employment, and local services and draw on existing social networks (Munshi, 2003; Zorlu & Mulder, 2008). Historical settlement patterns therefore provide a way to predict differences in local exposure to subsequent national immigration changes without relying solely on contemporaneous postcode-level settlement decisions. The instrument captures predicted exposure rather than actual immigration itself; its validity depends on whether this predicted exposure affects housing-price growth through immigration rather than through other channels.

Formally, for each immigrant income group $j \in \{\text{Total, Low, Middle-Low, Middle-High, High}\}$, the predicted immigrant exposure instrument $Z^j_{i,t-1}$ in postcode i and year $t-1$ is defined as:

$$Z^j_{i,t-1} = \sum_c \underbrace{\left(\frac{M_{c,i,t_0}}{M_{c,t_0}}\right)}_{\text{Share}} \times \underbrace{\left(\frac{\Delta M^j_{c,t-1}}{\text{Pop}_{i,t-2}}\right)}_{\text{National Shift}}, \tag{2}$$

where $t_0 = 1995$ denotes our baseline period, c indexes 21 distinct origin regions, and j denotes the income quartile.[5] The baseline share, $M_{c,i,t_0}/M_{c,t_0}$, represents the proportion of foreign-born individuals from origin c residing in postcode i in 1995 relative to the national total for origin c. The shift variable, $\Delta M^j_{c,t-1}$, measures the annual change in national immigrant arrivals from origin c in income group j at time $t-1$. Dividing the national inflow change, the shift, by the postcode population, $\text{Pop}_{i,t-2}$, puts the predicted exposure on a scale comparable to the endogenous immigration variable in equation (1). Thus, postcodes with larger historical shares of an origin group receive greater predicted exposure when national inflows from that origin increase.

The first-stage regression corresponding to baseline equation (1) is:

$$\frac{\Delta M^j_{i,t-1}}{\text{Pop}_{i,t-2}} = \theta^j Z^j_{i,t-1} + \boldsymbol{\gamma}' X_{i,t-2} + \boldsymbol{\theta}'(W_i \times \delta_t) + \sigma_i + \delta_t + \nu^j_{i,t-1}. \tag{3}$$

For $Z^j_{i,t-1}$ to be a valid instrument, two conditions must be satisfied: instrument relevance and conditional exclusion. First, the instrument must be sufficiently correlated with local immigrant inflows. Second, conditional on the controls and fixed effects, the instrument must affect local house price growth only through its effect on immigrant inflows. Relevance requires that

[5]Origin countries are aggregated into 21 regions based on visa status (e.g., EU/EFTA), geographical proximity, and historical inflow volumes. Major origin groups with large inflow volumes—such as Turkey, Indonesia, Germany, and Poland—are maintained as standalone categories to prevent small-sample noise from contaminating baseline shares.

historical 1995 location shares strongly predict subsequent local immigrant inflows when interacted with aggregate national shifts. This mechanism is motivated by the role of co-ethnic networks in facilitating job search, housing search, and social integration. Previous studies document the importance of immigrant networks in shaping location choices (Munshi, 2003) and provide evidence of co-ethnic settlement patterns in the Netherlands (Zorlu & Mulder, 2008). Across all baseline specifications, we assess first-stage strength using the Kleibergen–Paap r_k Wald F-statistic.

A further implication of the instrument is that new arrivals across income categories may follow established co-national settlement networks.

We examine whether the historical spatial concentration of immigrant groups is primarily organized by origin rather than income category. Specifically, we calculate the Herfindahl–Hirschman Index (HHI) for each origin group:

$$\text{HHI}_c = \sum_i \left(\frac{M_{c,i}}{M_i} \right)^2, \tag{4}$$

where $M_{c,i}$ denotes the stock of origin c in postcode i and M_i is the total foreign-born population in i. Table D.4 reports this descriptive evidence. The concentration patterns help characterize the spatial distribution of origin groups, although they do not, by themselves, establish the validity of the instrument.

Following the shift-share identification framework of Goldsmith-Pinkham, Sorkin, and Swift (2020), our identification strategy exploits the exogeneity of baseline settlement shares. Because spatial variation in exposure is driven by pre-existing 1995 settlement patterns across postcodes, the exclusion restriction requires that, conditional on controls and fixed effects, historical 1995 origin shares affect house price growth between 2006 and 2020 *solely* through their influence on instrument-predicted immigrant inflows. Relying on share exogeneity provides two distinct econometric advantages in our setting: it ensures consistency even if national origin-income shocks capture broader macroeconomic trends, and fixing shares at 1995 decouples instrument exposure from recent neighborhood gentrification trends (Borusyak, Hull, & Jaravel, 2025).

This exclusion restriction assumption may be violated if historical settlement patterns are correlated with local economic characteristics that subsequently affect housing demand independently of immigration. For example, postcodes with larger historical shares of a particular origin group may be more exposed to certain industries or labor-market developments. If those industries subsequently expand or contract, local employment, income, and housing demand may change even apart from the new immigration predicted by the instrument.

Our specification addresses several potential sources of confounding. First, postcode fixed effects (σ_i) absorb time-invariant differences across locations, while year fixed effects (δ_t) ac-

count for shocks common to all postcodes in a given year. Second, interactions between baseline house prices and baseline income and year fixed effects ($W_i \times \delta_t$) allow price trajectories to differ flexibly with initial local conditions. Third, lagged, time-varying controls ($X_{i,t-2}$) further account for observed changes in local unemployment, housing stock, and working-age population. These controls reduce the scope for certain alternative explanations but cannot eliminate every potential time-varying local shock.

We assess the identifying assumptions using several complementary exercises. First, we conduct a pre-trend falsification test to examine whether future values of the instrument predict current house price growth. The absence of significant leading effects would be consistent with the identifying assumptions, although it would not prove the exclusion restriction. Second, to eliminate potential own-observation feedback bias—where a postcode's own inflows might contaminate the aggregate national shift—our shift-share instrument is constructed using a leave-one-out formulation that explicitly excludes postcode i's own contribution.

We also examine whether the predicted shift-share instrument is systematically related to observed time-varying local economic characteristics. Following Goldsmith-Pinkham, Sorkin, and Swift (2020), we regress the predicted instrument for each immigrant category on changes in local housing supply, working-age population, and unemployment, including postcode and year fixed effects. Table 3 reports the results. Across all income groups j, the shift-share predicted inflows exhibit near-zero, statistically negligible correlations with local housing supply changes and working-age population dynamics. Similarly, controlling for fixed effects attenuates the relationship between the instrument and local unemployment rates toward zero. Furthermore, the within-R^2 values remain low across all specifications (0.070 to 0.122). This low explanatory power indicates that observable time-varying local economic drivers explain very little of the spatial exposure generated by the instrument. Combined with our postcode and year fixed effects, these balance results confirm that historical 1995 settlement patterns do not systematically track contemporaneous local economic shocks, supporting the conditional exogeneity of the shift-share instrument.

Finally, we assess whether our income-specific results are driven by the income classification. We tabulate the income distribution of immigrants within each major origin region in Table D.6 and find that immigrants from the same broad origin region are distributed across multiple income categories, and the degree of dispersion differs substantially across origins. We also verify that using fixed national income thresholds rather than municipal-relative income ranks yields similar estimates. This robustness check reduces concern that the results depend on the use of municipality-relative income classification.

Table 3: Balance Test: Regression of Shift-Share Instrument on Local Time-Varying Controls

	Dependent Variable: Shift-Share Instrument ($Z^{j}_{i,t-1}$)									
	All		High Income		Mid-High Income		Mid-Low Income		Low Income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Lagged Unemployment Rate	-0.008*	-0.009	-0.002	0.001	-0.006**	-0.002	-0.018**	-0.006	0.018**	-0.002
	(0.005)	(0.007)	(0.001)	(0.000)	(0.003)	(0.001)	(0.008)	(0.004)	(0.008)	(0.003)
Lagged Housing Stock Change	0.000	0.000**	0.000*	0.000*	0.000	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Lagged Workage Pop Change	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000***	0.000**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
FE: Year	X	X	X	X	X	X	X	X	X	X
FE: PC4	X	X	X	X	X	X	X	X	X	X
Observations	20,868	20,868	20,868	20,868	20,868	20,868	20,868	20,868	20,868	20,868
R^2 Within	0.082	0.081	0.071	0.073	0.089	0.091	0.121	0.118	0.122	0.121

Notes: Standard errors clustered at the municipality level in parentheses. The dependent variable in each column pair is the predicted shift-share instrument ($Z^{j}_{i,t-1}$) for the respective income tier j. All models include year and four-digit postcode fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 Results

We first show that the shift-share instrumental variable strategy yields a strong and relevant instrument. After that, we discuss the estimation results for the main outcomes of the tax assessment value (WOZ value) and the repeat-sales price index.

Table E.8 shows the estimation results for the first stage, using 1995 shares interacted with national-level year-on-year change of the stock of immigrants by region of origin and income type. Using the shifts per income type, we build an instrument for every income type of immigrants. This instrument is a strong predictor of the contemporaneous year-on-year change of the stock of immigrants per neighborhood, as indicated by a positive and statistically significant coefficients across all specifications. For example, a 1-percentage-point increase in the predicted inflow of high-income immigrants leads to an actual increase of 0.828 percentage points relative to neighborhood population.

The R^2 reflects a substantial fraction of explained variation. It is between 0.14 and 0.34, depending on the type of immigrant. Additionally, the first-stage F-statistic is very high, above 100. We also present the KP F-statistics (Kleibergen and Paap (2006) F-statistic) which account for clustered standard errors and test for weak instruments. While the majority of these F-statistics exceed the conventional threshold of 10 (ranging from 22.6 to 447.9), the middle-high income subgroup exhibits a lower KP F-statistic of 8.8, pointing to potential weak instrument concerns for that specific category. Overall, however, the results confirm a strong and significant instrument for the broad aggregate and most immigrant income tiers.

Table 5 and Table 6 report second-stage IV estimates of the effect of immigration on tax assess-

Table 4: First stage coefficients

	All migrants (1)	High income (2)	Middle-High income (3)	Middle-Low income (4)	Low income (5)
$\frac{\Delta M^{j}_{i,t-1}}{Pop_{i,t-2}}$	0.697***	0.889***	0.300**	0.713***	0.828***
	(0.106)	(0.187)	(0.101)	(0.042)	(0.039)
FE: Year	x	x	x	x	x
FE: PC4	x	x	x	x	x
N	30,048	30,048	30,048	30,048	30,048
Adj. R^2	0.278	0.343	0.143	0.273	0.243
IV F statistic	373.5	729.9	101.5	2089.9	420.7
KP F statistic	43.0	22.6	8.8	287.3	447.9

Note: First stage, as in Equation (3), with an income-type specific SSIV variable, using 1995 shares. One column per income-range of immigrant. Outcomes measured in 2007 - 2020. Standard errors in parentheses, clustered by municipality. The regression is weighted by the postcode-level population in 2006. "IV F" statistic reports the Cragg-Donald F-statistic under homoskedasticity. "KP F" reports the Wald statistic for weak instruments (Kleibergen-Paap rk statistic) accounting for clustered standard errors. Table E.7 shows that the first stage coefficients hardly change when including controls. Statistical significance level: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Source: CBS Microdata

ment values, measured by the WOZ administrative valuation, and on the Case-Shiller repeat-sales index respectively. Identifying variation is driven by a shift-share instrument that isolates exogenous changes in immigrant inflows via historical settlement patterns interacted with national sending-country flows. All specifications include year and 4-digit postcode fixed effects; even-numbered columns add time-varying local controls for unemployment, housing stock, and working-age population.

The main takeaway from Table 5 is that pooling all immigrants masks substantial heterogeneity across income tiers. While aggregate estimates remain imprecise, decomposing inflows by income group reveals clear, statistically significant impacts at the lower end of the income distribution. Middle-low income inflows generate a statistically significant positive effect on the WOZ value. In the baseline specification (column 7), the estimated effect evaluated at the mean regressor (0.0005) implies a 0.17 to 0.20 percentage point increase in tax assessment value—representing roughly 10% to 12% of the sample average annual property appreciation (0.0169). Conversely, low-income immigrant inflows exert a negative and statistically significant impact on house prices (column 9), although the estimate is no longer statistically significant when additional controls are included (column 10). Evaluated at the mean regressor (0.0008), the implied reduction of 0.0005 to 0.0006 represents a 3% to 4% drag relative to mean property appreciation. This negative elasticity is consistent with neighborhood filtering dynamics or changes in perceived local public amenities (e.g., school composition or public safety concerns) that lower the willingness to pay among incumbent residents. In contrast, the broader and higher-income sam-

Table 5: Second stage coefficients for tax assessment value (WOZ)

	Dependent variable: tax assessment value									
	All		High income		Middle-High income		Middle-Low income		Low income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\widehat{\frac{\Delta M^j_{i,t-1}}{Pop_{i,t-2}}}$	1.680	1.863	1.016	-1.321	9.560	5.543	4.012*	3.360	-0.597*	-0.735
	(2.263)	(3.137)	(10.241)	(9.364)	(17.004)	(18.314)	(2.245)	(3.096)	(0.308)	(1.834)
FE: Year	x	x	x	x	x	x	x	x	x	x
FE: PC4	x	x	x	x	x	x	x	x	x	x
Controls:		x		x		x		x		x
N	30,048	20,706	30,048	20,706	30,048	20,706	30,048	20,706	30,048	20,706
Avg outcome	0.0169	0.0169	0.0169	0.0169	0.0169	0.0169	0.0169	0.0169	0.0169	0.0169
Avg regressor	0.0021	0.0021	0.0003	0.0003	0.0004	0.0004	0.0005	0.0005	0.0008	0.0008
SD regressor	0.0064	0.0064	0.0015	0.0015	0.0021	0.0021	0.0022	0.0022	0.0049	0.0049
KP F statistic	12.9	12.9	15.4	15.4	5.2	5.2	148.4	148.4	107.1	107.1
Impact at mean regressor	0.0035	0.0039	0.0003	-0.0004	0.0038	0.0022	0.0020	0.0017	-0.0005	-0.0006

Note: Second stage coefficients, as in Equation (1), with a income-type specific SSIV variable, using 1995 shares. One column per income-range of immigrant. Controls include the year-on-year change on the unemployment rate, housing stock and total population. The outcome is measured in 2006 - 2020. Standard errors are clustered by municipality. Statistical significance level: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Source: CBS Microdata

ples yield less definitive conclusions. In the pooled sample (columns 1–2), the increase in the overall immigrant share produces a modest, positive point estimate of 0.35 in the baseline and 0.39 with controls at mean, though neither reaches statistical significance. Specifically, estimates for high-income (columns 3–4) and middle-high-income immigrants (columns 5–6) are large in magnitude but accompanied by very wide confidence intervals. Consequently, while point estimates for upper-income tiers lean positive, we cannot draw statistically reliable conclusions for these groups.

Across all specifications, adding time-varying municipality controls leaves the point estimates qualitatively stable, supporting the conditional exogeneity of the shift-share instrument.

Table 6 presents estimates using the Case-Shiller repeat-sales price index as the dependent variable. We treat these as a validation exercise rather than a primary result. The repeat-sales index is constructed from repeated property-level transactions, making it a market-based, transaction-price measure of house price growth that is free from the administrative smoothing inherent in the WOZ valuation. In principle, this makes it a sharper signal of true market conditions. In practice, however, it is subject to two important limitations in our setting. First, because it requires at least two transactions on the same property, it is based on a selected subsample of properties, and the selection may be correlated with neighborhood change — precisely the margins we are studying. Second, as shown in Table 2 and Appendix Figure D.2, the index exhibits substantially greater cross-neighborhood dispersion and lower average year-on-year variation than the WOZ measure, generating less statistical power.

The repeat-sales results broadly validate the qualitative pattern from the WOZ assessment

Table 6: Second stage coefficients for repeat-sales index

	Dependent variable: change in repeat sales index									
	All		High income		Middle-High income		Middle-Low income		Low income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\widehat{\frac{\Delta M^j_{i,t-1}}{Pop_{i,t-2}}}$	0.972**	-0.269	-1.737	-3.878***	4.175	0.448	1.731***	1.513	0.064	-1.558
	(0.416)	(0.478)	(1.637)	(1.206)	(3.505)	(6.441)	(0.512)	(1.424)	(0.322)	(1.972)
FE: Year	x	x	x	x	x	x	x	x	x	x
FE: PC4	x	x	x	x	x	x	x	x	x	x
Controls:		x		x		x		x		x
N	30,518	20,774	30,518	20,774	30,518	20,774	30,518	20,774	30,518	20,774
Avg outcome	0.0107	0.0107	0.0107	0.0107	0.0107	0.0107	0.0107	0.0107	0.0107	0.0107
Avg regressor	0.0021	0.0021	0.0003	0.0003	0.0004	0.0004	0.0005	0.0005	0.0008	0.0008
SD regressor	0.0064	0.0064	0.0015	0.0015	0.0021	0.0021	0.0022	0.0022	0.0049	0.0049
KP F statistic	12.9	12.9	15.4	15.4	5.2	5.2	148.4	148.4	107.1	107.1
Impact at mean regressor	0.0020	-0.0006	-0.0005	-0.0012	0.0017	0.0002	0.0009	0.0008	0.0001	-0.0013

Note: Second stage coefficients, as in Equation (1), with a income-type specific SSIV variable, using 1995 shares. One column per income-range of immigrant. Controls include the year-on-year change on the unemployment rate, housing stock and total population. The outcome is measured in 2006 - 2020. Standard errors in parentheses, clustered by municipality. Statistical significance level: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Source: CBS Microdata

analysis while highlighting key nuances specific to market transaction dynamics. Middle-low-income immigrant inflows again generate a strong and precisely estimated positive effect in the baseline specification ($\beta = 1.731$, column 7), which remains economically meaningful when adding controls ($\beta = 1.513$, column 8). Evaluated at the mean regressor, this represents an appreciation of 0.0008 to 0.0009—accounting for roughly 7%8% of average annual repeat-sales growth. Low-income immigrant inflows similarly align with the WOZ pattern, displaying a negative point estimate once controls are included (column 10), translating to an impact at the mean regressor of -0.0013 (a 12% drag relative to average annual price appreciation).

Two notable departures from the WOZ valuations are observable. The pooled estimate in column (1) is positive and statistically significant ($\beta = 0.972$), but flips sign to -0.269 upon the inclusion of time-varying controls (column 2). The high-income effect was positive but imprecisely estimated in the WOZ baseline, the repeat-sales index reveals a negative and highly statistically significant coefficient once controls are included (column 4), implying an impact at the mean regressor of -0.0012. These sign reversals suggest that year-on-year variation in transaction-based repeat-sales indices carries a **heightened macroeconomic and cyclical sensitivity that is correlated with aggregate and high-tier immigration flows.** Rather than contradicting our main findings, these differences reflect a known property of repeat-sales indices in settings with thin or cyclically sensitive transaction volumes, where market transactions react faster to cyclical shifts than administrative valuations. Finally, the estimates remain consistent across both valuation measures for middle-low income inflows consistently support local property appreciation.

5.1 Robustness

In this section, we demonstrate that our baseline estimates are robust to our sample selection criteria and examine the performance of alternative shift-share instrumental variable (SSIV) formulations and spatial aggregation levels.

Table 7 presents the second-stage results across these sensitivity checks, reporting standard errors clustered at the municipality level.

Column (2) evaluates sensitivity to the postcode inclusion threshold outlined in Table 1. While our baseline sample restricts attention to postcodes above the 30th percentile of the 1995 housing stock distribution, column (2) relaxes this threshold to the 20th percentile, incorporating smaller and lower-density postcodes. Because our regressions are weighted by the postcode population in 2006, the estimated parameters are virtually identical to the baseline in column (1): the estimate for middle-low-income immigrants remains 4.014, while the low-income estimate remains −0.600. This confirms that our primary selection criterion does not induce sample selection bias.

Column (3) examines the effect of aggregating the data from the 4-digit postcode level up to the municipality level. As expected, aggregating to whole municipalities reduces the sample size from 30,048 to 4,365 observations and averages out sub-neighborhood variation. At this broader spatial scale, point estimates attenuate toward zero across all income groups: the pooled estimate is −1.089 (SE = 3.313), middle-low income inflows attenuate to 0.887 (SE = 5.301), and low-income inflows yield −0.384 (SE = 0.418).

Columns (4) and (5) test the sensitivity of our IV strategy to variations in the construction of the shift-share instrument. Column (4) refines the shift component by replacing broader regional sending-area aggregates with disaggregated, country-level origin flows. The core pattern of income-based heterogeneity persists strongly under country-level shifts. In particular, the positive impact for middle-low-income immigrants remains large and becomes highly statistically significant ($\beta = 6.645$), while low-income inflows maintain a negative point estimate ($\beta = -1.676$). Column (5) replaces our baseline 1995 settlement shares with a rolling 10-year lagged share structure. Under this alternative specification, point estimates across all income groups attenuate and lose statistical significance (e.g., j = middle-low yields −0.594, SE = 3.484). This divergence highlights a key structural feature of the Dutch housing context: using a 10-year rolling lag introduces endogenous spatial realignment driven by short-run housing market cycles, whereas fixing initial shares to a base year isolates long-run settlement patterns that are genuinely exogenous to subsequent local price shocks.

Overall, these checks confirm that our primary baseline findings are not driven by arbitrary sampling cut-offs, demonstrate that disaggregating shifts to individual sending countries strengthens our baseline results, and highlight the empirical necessity of analyzing housing mar-

Table 7: Robustness of SSIV

	Dependent variable: change in tax assessment value				
$\frac{\widehat{\Delta M^j_{i,t-1}}}{Pop_{i,t-2}}$	Baseline (1)	Sample 20th Pct (2)	Municipality level (3)	Country-level shifts (4)	10y-lagged shares (5)
j = all	1.680	1.676	-1.089	-0.071	-0.165
	(2.263)	(2.263)	(3.313)	(2.346)	(1.972)
j = high income	1.016	1.006	-9.031	3.333	-7.054
	(10.241)	(10.239)	(8.169)	(6.396)	(6.279)
j = midhigh income	9.560	9.577	-13.939	8.083	-7.355
	(17.004)	(17.049)	(19.054)	(8.311)	(16.470)
j = Middle-Low income	4.012*	4.014*	0.887	6.645***	-0.594
	(2.245)	(2.247)	(5.301)	(0.349)	(3.484)
j = low income	-0.597*	-0.600*	-0.384	-1.676	0.895
	(0.308)	(0.308)	(0.418)	(2.788)	(1.052)
FE: Year	x	x	x	x	x
FE: PC4	x	x	x	x	x
x	x	x	x	x	
N	30,048	29,998	4,365	24,555	30,062

Note: Second stage coefficients, as in Equation (1), with a income-type specific SSIV variable. One column per sample, one row per regression. The outcome is measured in 2006 - 2020. Standard errors in parentheses, clustered by municipality. Column (2) is estimated on the a different sample of postcodes, columns (4) and (5) present a variation in the SSIV. Statistical significance level: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Source: CBS Microdata

ket adjustments at fine spatial scales.

As an additional robustness check, we verify that our results are not confounded by own-observation feedback bias. This mechanical endogeneity can arise in shift-share frameworks if a large or rapidly growing postcode i contributes significantly to the national total, causing local unobserved shocks to contaminate the national shift component. To rule this out, we re-estimate our baseline models using a leave-one-out shift-share instrument ($\Delta M^j_{c,-i,t-1}$), which explicitly subtracts postcode i's own inflows from the national origin-specific aggregate. The second-stage coefficients remain virtually identical to our baseline estimates across all income groups, confirming that local feedback effects do not bias our main results. Finally, we test whether our findings are sensitive to how immigrant income tiers are defined. While our baseline specification relies on municipality-relative income ranks, we re-classify foreign-born arrivals using fixed national income thresholds. Using national-level income cut-offs yields point estimates that remain close to our baseline coefficients, demonstrating that our main results capture structural income dynamics rather than artifactual local ranking shifts.

5.2 Heterogeneity by areas

Next, we turn to heterogeneity by sub-samples of geographic areas. The baseline sample contains neighborhoods across the entire country. However, housing and immigration dynamics may differ by region, for example in dense urban centers versus less densely populated areas. We split the sample into two complementary sets: Firstly, postcodes located in municipalities with at least 100,000 inhabitants in 2021, reflecting more urban and densely populated areas ($N = 482$ postcodes across 18 municipalities). In what follows, we refer to municipalities with more than $100,000$ inhabitants as "urban" and those below this threshold as "less urban." Secondly, we distinguish the four largest cities (Amsterdam, Rotterdam, The Hague, and Utrecht). This distinction follows Hekwolter, Nijskens, and Heeringa (2017), who note that housing markets in major metropolitan hubs experience distinct structural dynamics compared to the rest of the country.

We present descriptive statistics for urban and less urban municipalities in Appendix Table D.2 and highlight the main distinctions here. On average, postcodes in less urban municipalities contain a smaller population and fewer immigrants, resulting in a substantially lower immigrant share. Specifically, less urban postcodes contain an average of 4,742 inhabitants, compared to 7,272 inhabitants and 6,226 dwellings per postcode in urban municipalities. The immigrant share averages 27% in urban municipalities, but only 13% in less urban postcodes. While the housing stock is smaller in less densely populated postcodes, average household size is larger, and the share of owner-occupied units is significantly higher at 62% on average, compared to 43% in urban municipalities.

We report the second-stage heterogeneity results for our leading property value measure (the WOZ value) across these sub-samples in Table 8. The estimates reveal stark geographic divergence: the positive property value effects of immigration are overwhelmingly driven by less densely populated areas (municipalities with $< 100,000$ inhabitants and outside the "Big 4" cities).

Specifically, for the set of postcodes outside the four big cities (column 5) and in less urban municipalities (column 3), the estimated coefficients are large, positive, and statistically significant. For instance, the baseline effect for all migrants is 4.711 outside the Big 4 and 6.041 in smaller municipalities. When broken down by income tiers, high-income immigrant inflows in non-Big-4 areas drive massive positive responses ($\beta = 16.982$), and middle-low income inflows also show robust positive valuation effects ($\beta = 5.698$). In sharp contrast, second-stage estimates for the year-on-year change in WOZ value are statistically indistinguishable from zero across all immigrant tiers in the four big cities and large urban municipalities ($> 100k$).

Potential explanations for this striking urban-rural heterogeneity involve market tightness, demographic salience, and housing structure. Hekwolter, Nijskens, and Heeringa (2017) char-

Table 8: Second stage coefficients for different samples

$\widehat{\frac{\Delta M^{j}_{i,t-1}}{Pop_{i,t-2}}}$	*Dependent variable: Lead of WOZ value*				
		Municipalities		Cities	
	Baseline (1)	>100k (2)	< 100k (3)	Big 4 (4)	< Big 4 (5)
j = all	1.188	0.215	6.041***	0.253	4.711***
	(1.973)	(2.406)	(1.455)	(2.419)	(0.781)
j = high income	8.309	-4.706	16.691***	-8.248	16.982***
		(11.598)	(6.516)	(11.887)	(3.146)
j = midhigh income	15.384	4.267	15.273***	-4.136	30.908
	(9.661)	(41.373)	(5.758)	(38.638)	(26.347)
j = Middle-Low income	7.484***	2.636	5.758	1.251	5.698***
	(0.444)	(2.995)	(0.968)	(3.606)	(0.507)
j = low income	-1.810	-0.653	0.834	0.083	-0.633
	(2.078)	(0.365)	(0.887)	(0.484)	(0.517)
FE: Year	x	x	x	x	x
FE: PC4	x	x	x	x	x
N		6,160	21,999	1,843	26,316
N Postcodes	2,349	493	1,724	154	2,063
N Municipalities	341	18	322	4	337

Note: Second stage coefficients, as in Equation (1), with an income-type specific SSIV variable, using 1995 shares. One column per sample. The outcome is measured in 2006 - 2020. Standard errors in parentheses, clustered by municipality. Statistical significance level: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Source: CBS Microdata

acterize housing markets in large Dutch cities as heavily "overheated," marked by rapid transactions above asking prices and structural supply bottlenecks. In such tightly constrained environments, prices are largely driven by macro-financial conditions, bidding wars, and structural shortages rather than marginal demographic or immigrant shocks. This is reflected in the descriptive data showing an average annual WOZ growth rate of 3.6% in urban municipalities versus 2% in less urban areas. Conversely, less densely populated areas offer more elastic margins where local demographic inflows can directly translate into measurable price adjustments.

Immigration is far less common in smaller, less urban areas, making demographic changes more visible and economically salient to incumbent residents. Survey data from the World Values Survey (Table E.10) and findings by Card, Dustmann, and Preston (2005) using the European Social Survey indicate that residents in smaller towns report lower baseline exposure and lower stated comfort levels regarding neighbors of different nationalities, languages, or races

compared to residents in major cities > 500,000 inhabitants. While these surveys are indicative rather than directly matched to our exact postcode sample, they suggest that residential sorting and local market capitalization of immigrant inflows may interact with regional differences in social composition and openness.

The predominance of owner-occupied housing in less urban areas (62% vs. 43% in urban centers) shapes how local housing markets clear. In owner-dominated submarkets, price adjustments via the WOZ tax assessment value reflect transaction-based pricing and homeownership demand channels more directly than in rental-heavy urban neighborhoods, where rent controls and institutional landlords may mute official valuation responses.

5.3 Results for the medium term

To examine whether the effects identified in Table 5 persist or reverse over time, Table E.9 replaces the dependent variable with the three-year change of the tax assessment value, while keeping the regressor and shift-share instrument unchanged. This specification addresses a central dynamic question: does a one-year immigrant inflow shock in year $t-1$ continue to capitalize into property values through $t+3$, or does the initial price response unwind over a medium-term horizon? The results indicate that the estimated medium-term responses vary across specifications and immigrant income groups. In the pooled specification, the controlled model yields a negative and statistically significant coefficient of (-14.397, column 2). Evaluated at the mean regressor value of (0.0021), this corresponds to an estimated effect of (-0.0302). This result is consistent with weaker property value growth over the medium-term horizon, although it does not by itself establish that earlier price increases are reversed.

Decomposing the inflows by income group uncovers important compositional nuances across the distribution. In contrast to their positive contemporaneous effect, middle-low income inflows exhibit negative point estimates at the three-year horizon. The baseline coefficient of -10.127 (column 7) and the controlled coefficient of -15.550 (column 8) translate to mean impacts of -0.0051 and -0.0078, respectively. These estimates suggest that the positive short-run association does not necessarily persist over the medium term. However, they do not establish whether this pattern reflects a reversal in house-price growth, changes in local housing demand, or other forms of adjustment.

Low-income inflows show positive point estimates at the three-year horizon ($\beta = 3.797$, in column 9; $\beta = 2.119$ in column 10), corresponding to mean impacts of 0.0030 and 0.0017. High-income and middle-high income specifications yield large point estimates in both directions, but these estimates are accompanied by wide standard errors, precluding definitive statistical conclusions.

Taken together, the differences between the one-year and three-year estimates suggest that

Table 9: Effect of a change of immigrant share on three-year change in tax assessment value

	Dependent variable: 3-year change tax assessment value									
	All		High income		Middle-High income		Middle-Low income		Low income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\widehat{\frac{\Delta M^{j}_{i,t-1}}{Pop_{i,t-2}}}$	-1.107	-14.397*	8.078	-19.120	63.404	-91.060	-10.127	-15.550	3.797*	2.119
	(4.914)	(7.789)	(22.760)	(25.537)	(55.626)	(64.587)	(11.253)	(16.087)	(2.019)	(5.797)
FE: Year	x	x	x	x	x	x	x	x	x	x
FE: PC4	x	x	x	x	x	x	x	x	x	x
Controls:		x		x		x		x		x
N	25,427	17,178	25,427	17,178	25,427	17,178	25,427	17,178	25,427	17,178
Avg outcome	0.0388	0.0388	0.0388	0.0388	0.0388	0.0388	0.0388	0.0388	0.0388	0.0388
Avg regressor	0.0021	0.0021	0.0003	0.0003	0.0004	0.0004	0.0005	0.0005	0.0008	0.0008
SD regressor	0.0064	0.0064	0.0015	0.0015	0.0021	0.0021	0.0022	0.0022	0.0049	0.0049
Impact at mean regressor	-0.0023	-0.0302	0.0024	-0.0057	0.0254	-0.0364	-0.0051	-0.0078	0.0030	0.0017

Note: Outcome: year-on-year change in the three-year change of WOZ value per four-digit postcode. Second stage coefficients, as in Equation (1), with an income-type specific SSIV variable, using 1995 shares. The outcome is measured in 2006 - 2020. Standard errors in parentheses, clustered by municipality. Statistical significance level: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Source: CBS Microdata

the estimated effects of immigration on local property values may evolve over time and vary by the income composition of inflows. In particular, the negative three-year estimates for middle-low-income inflows contrast with their positive contemporaneous estimates, while the low-income estimates become positive at the three-year horizon despite negative short-run estimates. These patterns are consistent with medium-term adjustment in local housing markets, but they do not establish a specific mechanism such as incumbent displacement, native outmigration, or neighborhood transition. Changes in the native population and other measures of residential mobility may provide additional evidence on these potential channels.

5.4 Natives

To investigate whether the housing market impacts of immigrant inflows operate through residential sorting and native relocation, we examine postcode-level native population growth—defined as the annual change in the native population scaled by lagged native stock—as an alternative dependent variable. The results are reported in Table 10. The results reveal heterogeneity in native sorting that helps contextualize the house price findings. Middle-high-income immigrants induce the strongest negative native population response, with estimates ranging from -6.911 (column 5) in the baseline to -8.083 (column 6) with controls. Inflows of high-income immigrants similarly display a statistically significant negative effect ($\beta = -4.051$, column 4), as do middle-low-income immigrants in the baseline specification ($\beta = -1.646$, column 7).

Because high- and middle-high income inflows trigger substantial native out-migration or reduced native entry without altering housing prices, native relocation in upper-income tiers

Table 10: Second stage coefficients for change in natives

	Dependent variable: change in natives									
	All		High income		Middle-High income		Middle-Low income		Low income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\widehat{\frac{\Delta M^j_{i,t-1}}{Pop_{i,t-2}}}$	-0.567	-1.040*	-3.547**	-4.051**	-6.911***	-8.083**	-1.646***	-1.791	0.738***	1.502
	(0.407)	(0.560)	(1.611)	(1.701)	(1.981)	(3.354)	(0.569)	(1.088)	(0.147)	(0.922)
FE: Year	x	x	x	x	x	x	x	x	x	x
FE: PC4	x	x	x	x	x	x	x	x	x	x
Controls:		x		x		x		x		
N	30,736	20,868	30,736	20,868	30,736	20,868	30,736	20,868	30,736	20,868
Avg outcome	0.0033	0.0033	0.0033	0.0033	0.0033	0.0033	0.0033	0.0033	0.0033	0.0033
Avg regressor	0.0021	0.0021	0.0003	0.0003	0.0004	0.0004	0.0005	0.0005	0.0008	0.0008
SD regressor	0.0064	0.0064	0.0015	0.0015	0.0021	0.0021	0.0022	0.0022	0.0049	0.0049
Impact at mean regressor	-0.0012	-0.0022	-0.0011	-0.0012	-0.0028	-0.0032	-0.0008	-0.0009	0.0006	0.0012

Note: Second stage coefficients, as in Equation (1), with a income-type specific SSIV variable, using 1995 shares. One column per income-range of immigrant. Controls include the year-on-year change on the unemployment rate, housing stock and total population. The outcome is measured in 2006 - 2020. Standard errors in parentheses, clustered by municipality. Statistical significance level: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Source: CBS Microdata

appears to absorb the incoming housing demand, stabilizing local property valuations. In contrast, low-income immigrant inflows yield a positive coefficient in the baseline ($\beta = 0.738$, column 9), but this effect completely loses statistical significance once conditioning on time-varying controls ($\beta = 1.502$, column 10). The lack of a robust native migration response indicates that the negative property price impact observed for low-income inflows is not driven by physical native displacement or flight, but rather by negative amenity spillovers or downward neighborhood filtering. Finally, in the pooled sample (columns 1–2), the overall native growth estimate is negative ($\beta = -0.567$ in baseline; $\beta = -1.040$, with controls). Taken together, these findings confirm that while sorting dynamics absorb demand shocks in upper-income submarkets, price adjustments dominate in lower-income tiers.

6 Discussion and conclusion

This paper examines how immigration affects neighborhood-level property values in the Netherlands, distinguishing immigrant inflows according to their position in the municipality-level income distribution. Using comprehensive administrative data and a shift-share instrumental variable strategy, we investigate whether the effects of immigration differ across income groups, geographic areas, and time horizons. The results show that treating immigrants as a homogeneous group can conceal important differences in local housing-market responses.

Three findings emerge from the analysis. First, the contemporaneous relationship between immigration and property value growth varies by immigrant income. In the pooled specifica-

tion, an increase in the immigrant share is associated with higher year-on-year property value growth. However, this average effect masks substantial heterogeneity: the estimated effects are positive for immigrants in the upper income quartiles, whereas low-income immigrant inflows are associated with lower property value growth in the baseline results. These findings suggest that the economic composition of immigration matters for understanding local housing-market dynamics. Income may influence the types of housing households demand, their ability to compete for properties, and the neighborhoods in which additional demand is concentrated. Nevertheless, because income is measured relative to the municipality-level distribution, these categories should be interpreted as relative income groups rather than direct measures of wealth or housing purchasing power.

Second, the estimated effects differ markedly across geographic areas. The positive property value responses are concentrated in less urban municipalities and areas outside the four largest Dutch cities. In contrast, estimates for the largest cities and more urban municipalities are generally statistically indistinguishable from zero. One possible explanation is that housing-market conditions differ across these locations. In large cities, where housing demand is already high and supply constraints are substantial, marginal demographic changes may be less readily distinguishable from broader market dynamics. In less urban areas, where immigrant shares are generally lower and owner-occupied housing is more prevalent, changes in local demand or neighborhood composition may translate more visibly into assessed property values. Differences in housing-market structure, demographic composition, and local attitudes toward immigration may also contribute to the observed heterogeneity. These explanations are plausible interpretations of the results, however, rather than mechanisms directly identified by the empirical design.

Third, the medium-term estimates indicate that the short-run price responses do not necessarily persist. When property value changes are measured over a three-year horizon, the pooled estimate becomes negative in the specification that includes time-varying controls. The estimated medium-term effects also differ across income groups and are generally less precisely estimated. This pattern is consistent with the possibility that housing markets adjust over time through changes in residential location, household composition, and demand. We also examine changes in the native population. Although some specifications indicate declines in native population growth following particular immigrant inflows, these responses are not robust across all income groups and control specifications. Consequently, native out-migration remains a possible contributor to longer-term adjustment, but the evidence does not establish it as the principal mechanism behind the negative medium-term price estimates.

The findings contribute to the literature on immigration and housing by showing that estimated price effects depend on who arrives, where arrivals settle, and the time horizon over

which outcomes are measured. They also highlight the importance of distinguishing local property value responses from broader changes in housing affordability. An increase in assessed property values does not necessarily imply an equivalent increase in rents, nor does a decline in local price growth establish that housing has become more affordable for all households. The analysis therefore captures one important dimension of housing-market adjustment, rather than the full distributional consequences of immigration.

Several limitations should be considered when interpreting the results. Although the shift-share instrument exploits historical settlement patterns and national inflows to address the endogenous location choices of immigrants, the causal interpretation depends on the assumption that the instrument affects local property values only through the induced change in immigrant shares, conditional on the included controls and fixed effects. Historical settlement patterns may also be related to persistent local economic or demographic trends. In addition, some income-specific estimates are imprecise, and the strength of the first stage varies across groups. The results for individual income categories should therefore be interpreted with appropriate caution. Finally, property valuations and transaction-based price measures capture different aspects of the housing market, while the absence of population-wide rental price information limits what can be concluded about the effects on tenants.

Future research could extend this analysis by examining rental prices, housing tenure, and the distributional consequences for incumbent residents. and how these decisions interact with local housing demand. Further analysis of housing supply and local market conditions would help establish why price responses differ between urban and less urban areas.

Overall, the results demonstrate that the relationship between immigration and housing prices is heterogeneous rather than uniform. Income composition, geographic context, and the horizon of adjustment all matter for interpreting local property value changes. Accounting for these dimensions provides a more nuanced understanding of how immigration interacts with housing markets and cautions against drawing general conclusions from aggregate estimates alone.

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A Appendix

Table A.1: Literature overview, most to least recent

Authors	Country	Region	Type of migrant	Main result
Boje-Kovacs et al. (2024)	Denmark, 1987-2017	neighborhoods building unit	non-Western, refugees	0.3% ↓ house prices
Fernández-Huertas Moraga, Ferrer-i-Carbonell, and Saiz (2019)	Spain 1998-2008	neighborhood (census tracts, N = 28,523)	foreign-born, excl. children	0.33% ↑ native outflow; 50% ↑ construction
Mussa, Nwaogu, and Pozo (2017)	US, 2002 - 2012	MSA (N = 275)	foreign-born	0.8% ↑ house prices 0.8% ↑ rental prices 9.6% ↑ house prices in surrounding MSAs 1.6% ↑ rent in surrounding MSAs
Sá (2015)	UK, 2003-2010	local authorities (N =170)	foreign-born, 16-65 years	1.7% ↓ house prices; 0.05% ↑ native net out-migration
Gonzalez and Ortega (2013)	Spain, 2000-2010	provinces (N = 50)	foreign-born, working age	2% ↑ house prices; 1.2-1.5% ↑ construction of new houses
Accetturo et al. (2014)	Italy, 2003-2010	20 large cities & districts	foreign-born	0.5% ↑ house prices; 0.2% ↓ house prices within district; 0.6 ↑ outflow of natives
Akbari and Aydede (2012)	Canada 1996, 2001, 2006	Census Metropolitan Area (N = 258)	immigrants, recent & settled	0.1-0.12% ↑ house prices; from migrants arrived > 10 yrs ago
Saiz and Wachter (2011)	US, 1980, 1990, 2000	MSA (N = 67) Census tracks	foreign-born	0.134% ↓ native population; 0.678% ↓ non-Hispanic-white population 6% ↓ in housing values
Saiz (2003)	US, 1974 - 1983	MSA, Miami and Florida	Mariel boatlift	8-11% ↑ in units of low quality; 8% ↑ additional for units inhabited by low income Hispanic residents in 1979 than by non-Hispanic

B Data appendix

B.1 Postcode selection

In the following, we describe in detail the selection of postcodes. Using comprehensive data on population per postcode from 1995 - 2022, the goal is to construct a balanced panel of postcodes. The variables of interest for this study are the (change in) the stock of migrants and the stock of population. Firstly, 4,056 4-digit postcodes are consecutively available between 1995-2022. In the next steps, we exclude postcodes that ever have a registered population of 0 inhabitants, with the reference point being January 1st. Next, we exclude postcodes that exhibit an absolute growth rate of immigrants greater than 100% as they may be outliers in the analysis. The last restriction on the sample is the stock of inhabitants in 1995, the first year of observations. We exclude very sparsely population postcodes, as we are interested in changes in the demographic composition. Therefore, we present a sample selected on a minimum threshold of inhabitants, defined as the 10th, 20th and 30th percentile of the population distribution in 1995.

B.2 Individuals

We combine datasets on individuals, house transactions and values and neighborhoods from Statistics Netherlands (CBS) and Land Registry (Kadaster). These datasets cover the whole territory of the Netherlands. Demographic information is available for every resident in the Netherlands: date and place of birth, country of origin, gender, parent's dates and places of birth and country of origin. Immigrants living in the Netherlands for at least four months are required to register the address of residency at the Dutch municipality, together with the date of arrival and departure and the type of migration. There are four migration types: 1) Immigration, settlement from abroad; 2) Administrative inclusion, resettlement following a removal of the person without them ever leaving the Netherlands; 3) Emigration, removal of a person from the registry when departing abroad for at least 8 months; 4) Administrative discharge, removal from registry when the person is unreachable. We select immigrants type 1). The linked employer-employee dataset contains information on their monthly wage, gross and net, hours worked and dates of employment for regularly employed individuals. Due to the registration requirement, every individual is linked to their house address and dates of start and end of residence. If the house has been sold, for every sale the Land registry records transaction price, date, house value at the time of the transaction, recent house value, tenure type.

Characteristics of neighborhoods are reported in *Kerncijfers wijken en buurten* (KWB) for the year 1995, 1997, 1999, 2001, and from 2003 onwards. In these datasets, the characteristics are reported at municipality, district (*wijk*) and neighborhood (*buurt*) levels. We aggregate the data

of neighborhoods (*buurt*) to 4-digit postal code level. A postcode might contain more than one neighborhood.

B.3 Tax assessment value (WOZ)

To assess statutory residential property values (Waardering onroerende zaken, WOZ), Dutch municipalities employ automated valuation models (AVMs) that estimate the market value of unsold properties based on transaction prices of comparable local sales. Non-arm's-length transactions or sales occurring under non-standard market conditions are systematically excluded from the estimation sample. The econometric model incorporates key structural and locational attributes, including property type, building volume and floor size, plot size, construction year, physical location, and overall maintenance status. To maintain data accuracy, municipalities supplement market comparables with physical and remote verification checks. These include spatial audits using aerial imagery to detect structural alterations (e.g., extensions or dormers), physical neighborhood surveys, review of online real estate listings, cross-validation against Land Registry (Kadaster) plot records, and direct owner inquiries regarding recent internal renovations. Crucially, WOZ values are determined retrospectively with a one-year lag (e.g., the reference date for the 2022 WOZ valuation is January 1, 2021). Because unobserved internal property characteristics (e.g., interior finish quality) are not fully captured by administrative records, statutory WOZ valuations can deviate from realized transaction prices during periods of rapid market adjustment, underscoring the importance of pairing WOZ assessments with repeat-sales transaction indices.

C Repeat sales index

The repeat sales index represents the price appreciation in a given area over time. Using repeated sales of the same property allow to control for property-specific time-invariant characteristics. We obtain the repeat sales index following Bailey, Muth, and Nourse (1963). The model is given by:

$$r_{itt'} = \sum_{j=1}^{T} \beta_j x_j + u_{itt'} \tag{5}$$

where $r_{itt'}$ is the natural logarithm of the ratio of the final sales price in year t' and the initial sales price in year t for each transaction pair of dwelling i. The variable x_j takes value -1 if time j is the year of initial sale, 1 if j is the year of final sale, and 0 in all other periods. The OLS estimate $\hat{\beta}_j$ gives the log price aggregate index in year j.

To estimate the regression, we use the set of properties sold at least twice from 1995 to 2023.

This yields a number of transaction pairs for house i equal to the number of times the house has been sold minus 1.
First, we create dummy variables x_j, then the natural logarithm of the ratio of final and initial sales prices. Finally, we estimate the regression for each dwelling i. The coefficient $\hat{\beta}_j$ represents the house price index per 4-digit postcode.
Almost half of the dwellings sold in our sample have been transacted only once. Among the 4-digit postcodes in the 30 selected municipalities, there are 48 postcodes with houses sold only once between 1995 and 2023. Within postcodes there is a lot of heterogeneity in how many years pass between two transactions for the same house. The range goes from 0 to 34. On average across postcodes, the average sale time of a house is 6 years, with the 75th percentile being 8 years. Thus, houses sold once every 25 years are outliers.
The drawback of this methodology is that the variance increases with the interval of time between transactions. Hence, the OLS estimate is not efficient (Ghysels et al., 2013). The houses sold more than twice may not be representative of the whole sample, leading to a selection bias. A possible solution for this issue is offered in Case and Shiller, 1987. In their specification, they include a term to capture the imperfections of the housing market. Also, they have a component increasing with the length of the holding interval. Their estimation approach has three steps:

1. Estimate regression (5) and store residuals
2. Weighted list square regression of the residuals squared on a constant and on the time interval between transactions
3. GLS regression of (5), where the weights are the inverse of the square root of the fitted values from point 2.

Moreover, the repeat sales index does not consider what might determine the price of a property at a given point in time.

Then, we calculate the repeat sale index over time per postcode, using the Case and Shiller method. The steps are the following:

- Select a postcode and keep only the unique combinations of property identifier and sale date.
- Keep properties that are sold more than once If the subset for that postcode has more than 10 observations – this could imply that there could be few houses sold multiple times – proceed as follows for every year and postcode:
- Create a dummy variable per year from 1995-2023 and per property unit identifier

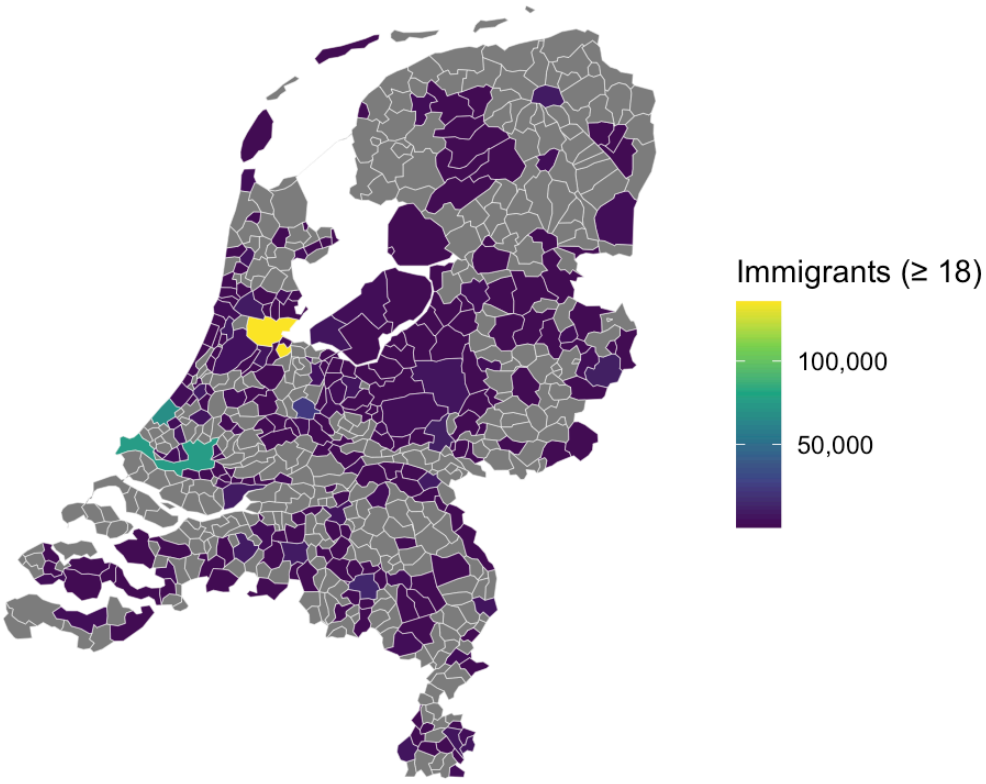
- Per unit, if the earliest sale date is equal to the dummy year, assign -1, this is because the first sale date should have -1. If the latest date is the same as the dummy year, assign 1. Otherwise 0.
- For houses with more than 2 sales date, per property identifier, calculate the following ratio: $log(\text{sale price}_t/\text{sale price}_{t-1})$
- Run the regression: $log(\text{sale price}_t/\text{sale price}_{t-1}) \sim$ dummy
  - Calculate the interval between sales year
  - Take the square of the residuals
  - Run the regression: squared residuals on time interval with constant
  - Store the fitted values
- For the third step: store the inverse of the square root of the fitted values: $\frac{1}{\sqrt{(fittedvalues)}}$
- Run the regression: ratio on all dummies per year with weights the inverse residuals, without constant.
- Save the exponential of the coefficients.

D Additional Statistics

Figure D.1 illustrates the distribution of immigrant shares across municipalities in the estimation sample in 1995. The distribution provides descriptive evidence on the extent of geographical variation in immigrant presence, which motivates our analysis of how local housing-price dynamics respond to immigrant inflows. The empirical strategy used to identify these effects is described in the next section.

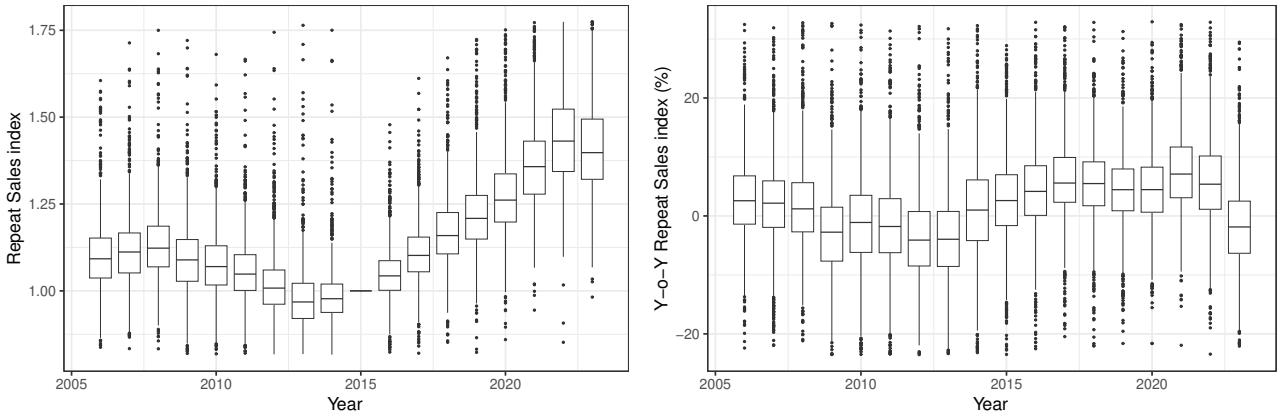
E Additional Results

Figure D.1: Number of immigrants in 1995 per municipality



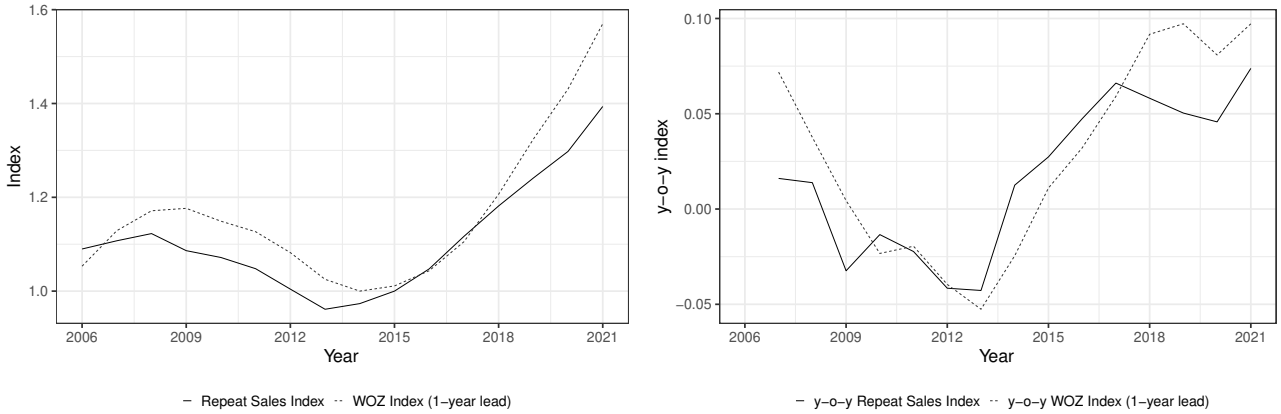
Notes: The map of the Netherlands shows the number of immigrants per municipality in 1995, in the sample selected.
Sources: CBS Microdata; *Kerncijfers Wijken en Buurten* (KWB).

Figure D.2: Levels and year-on-year (%) change of Repeat Sales Index across postcodes



Note: Levels of repeat sales index (LHS) and year-on-year change of repeat sales index (RHS) across years and postcodes. For every year, the boxplot shows the distribution of repeat sales index values across neighborhoods. Per year, the box shows the interquartile range (25th to 75th Percentile) and the line reflects the median. The whiskers reflect the range of values within median ±1.5× interquartile range. Dots reflect data points outside of the whiskers. Sample of 1,795 postcodes as outlined in Table 1.
Source: CBS Microdata, Kerncijfers Wijken en Buurten (KWB)

Figure D.3: Country aggregated index of WOZ value and Repeat Sales Index



Note: Country wide index of WOZ index and repeat sales index. Both series are a weighted mean across neighborhoods, weighted by the number of dwellings. Both are index in 2015. The left-hand side shows the level of the index and the right hand side shows the year-on-year growth. The correlation between national level year-on-year WOZ and year-on-year repeat sales index is 0.86. Sample of 1,795 postcodes as outlined in Table 1.
Source: CBS Microdata, Kerncijfers Wijken en Buurten (KWB)

Table D.2: Descriptive statistics for subsamples

	Sample: Urban Municipalities			
	Years	Mean	SD	N
Housing & Property Outcomes:				
WOZ growth rate ($\Delta \log P_{i,t}$)	2006–2021	0.0137	0.0515	1,709
Repeat-sales index growth	2006–2021	0.0092	0.1184	1,709
Demographics & Labor Market:				
Total population ($\text{Pop}_{i,t}$)	2006–2021	4,742.03	2,584.20	1,709
Unemployment rate	2006–2021	2.30%	1.03%	1,709
Average income per capita (€)	2009–2021	29,809.60	3,911.69	1,709
Immigrant Inflows ($\Delta M^{j}_{i,t-1}/\text{Pop}_{i,t-2}$):				
Pooled inflows (j = All)	2006–2021	0.00161	0.00514	1,709
High-income inflows (j = High)	2006–2021	0.00016	0.00111	1,709
Mid-high income inflows (j = Mid-High)	2006–2021	0.00045	0.00183	1,709
Mid-low income inflows (j = Mid-Low)	2006–2021	0.00038	0.00183	1,709
Low-income inflows (j = Low)	2006–2021	0.00063	0.00398	1,709
Postcode-year observations (N_{obs})		24,080		1,709

	Sample: Rural Municipalities			
	Years	Mean	SD	N
Housing & Property Outcomes:				
WOZ growth rate ($\Delta \log P_{i,t}$)	2006–2021	0.0256	0.0678	640
Repeat-sales index growth	2006–2021	0.0148	0.1478	640
Demographics & Labor Market:				
Total population ($\text{Pop}_{i,t}$)	2006–2021	5,374.37	3,408.57	640
Unemployment rate	2006–2021	2.53%	1.17%	640
Average income per capita (€)	2009–2021	30,443.30	5,924.16	640
Immigrant Inflows ($\Delta M^{j}_{i,t-1}/\text{Pop}_{i,t-2}$):				
Pooled inflows (j = All)	2006–2021	0.00402	0.00952	640
High-income inflows (j = High)	2006–2021	0.00079	0.00245	640
Mid-high income inflows (j = Mid-High)	2006–2021	0.00073	0.00297	640
Mid-low income inflows (j = Mid-Low)	2006–2021	0.00078	0.00292	640
Low-income inflows (j = Low)	2006–2021	0.00172	0.00705	640
Postcode-year observations (N_{obs})		9,005		640

Note: Descriptive statistics across 4-digit postcodes and years. Δ denotes the year-on-year percentage change. Share of rental and owner-occupied units is relative to the stock of housing. Migrant share, low-income and high-income share are relative to the population on Jan 1st in a given year. Share of low-income (high-income) population is defined as the number of households that belong to the bottom 40% (top 20%) of income in the entire country. Both share of low-income and high-income population are available from the KWB dataset and only indicated if there are at least 100 households by neighborhood.

Source: CBS Microdata, Kerncijfers Wijken en Buurten (KWB).

Table D.3: Groups used for SSIV

Group	Countries
Antilles	
China	
EFTA	Liechtenstein, Norway, Switzerland
GIPS	Greece, Italy, Portugal, Spain
Germany	
India	
Indonesia	
Middle East	Israel, Yemen, Iraq, Iran, Afghanistan, Jordan, Pakistan, Lebanon
Middle-Eastern Europe excl. Poland	Bulgaria, Estonia, Hungary, Latvia, Lithuania, Croatia, Romania, Slovenia, Slovakia, Czech Republic
Middle-South America	
Morocco	
Non-EU Europe	Iceland, Serbia, North Macedonia, San Marino, Russia, Andorra, Montenegro, Moldova, Belarus, Ukraine, Bosnia and Herzegovina, Albania, Monaco, Gibraltar, Isle of Man, Kosovo, Kazakhstan, Georgia, Azerbaijan, Armenia
North Africa	Tunisia, Syria, Egypt, Algeria
North EU	Austria, Belgium, Cyprus, Denmark, France, Finland, Great Britain, Ireland, Luxembourg, Malta, Sweden
Oceania, AUS, NZ	Australia, New Zealand
Poland	
Remaining Africa	African countries except for countries in category "North Africa"
Remaining Asia	
Suriname	
Turkey	
US, Canada	

Note: We aim to construct groups of origin countries that are similar in size by using the following criteria: visa status (i.e. grouping EFTA countries), geographic proximity, language proximity. Some countries are kept as their own category dues to large migration flows (i.e. Turkey, Indonesia, Germany, Poland).
Names of countries and continents refer to geographical entities as classified by CBS.

Table D.4: Herfindahl-Hirschmann Index for migrants by country of origin, 2006 - 2022

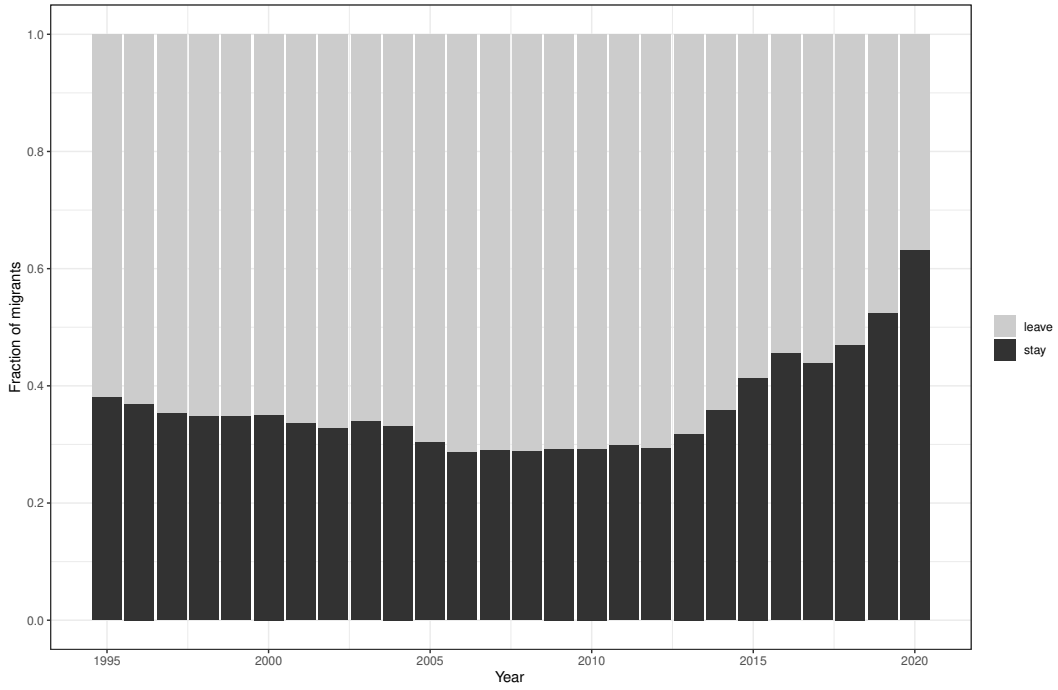
Origin group	HHI (all)	HHI (high income)	HHI (low income)	Mean (all)	SD (all)	Total inflow (all)
Antilles	0.08	0.07	0.08	73	80	665,062
China	0.13	0.14	0.17	51	60	342,583
EFTA	0.69	0.81	0.79	21	12	13,855
GIPS	0.05	0.07	0.05	64	67	902,010
Germany	0.05	0.05	0.06	59	79	879,345
India	0.08	0.14	0.04	69	85	646,985
Indonesia	0.10	0.10	0.13	24	18	124,236
Morocco	0.08	0.07	0.09	141	184	1,499,972
Middle East	0.04	0.04	0.03	68	82	949,480
Middle-Eastern Europe excl. Poland	0.05	0.06	0.06	61	53	617,345
Middle-South America	0.06	0.07	0.06	47	60	1,106,404
Non-EU Europe	0.03	0.04	0.02	52	45	997,182
North Africa	0.37	0.21	0.51	46	94	1,505,925
North EU	0.04	0.06	0.04	71	53	251,886
Oceania, Australia, NZ	0.56	0.67	0.59	29	25	30,510
Poland	0.05	0.04	0.05	65	87	1,108,929
Remaining Africa	0.06	0.05	0.06	78	120	1,387,515
Remaining Asia	0.05	0.14	0.05	53	52	737,796
Suriname	0.12	0.12	0.11	137	229	1,395,224
Turkey	0.07	0.06	0.08	149	184	1,762,391
US/Canada	0.22	0.29	0.25	50	69	218,718

Table D.5: Descriptive statistics for migration and housing

	25th Pct.	Mean	Median	75th Pct.	N
Age at immigration	23	30,9	28	35,6	2,674,225
Number of migration spells	1	1,14	1	1	2,674,225
Duration of migration spells (days)	292	1141	653	1,431	1,400,039
Number of address spells	x	x	x	x	2,674,225
Duration at address (days)	161	727.7	381	880	5,248,119

Note: Number and moments of immigration and address spells for migrants. Duration is conditional on completing a spell.
Sample: All migrants arriving between 1995 and 2021, age ≥ 18 years upon arrival.
Source: CBS Microdata

Figure D.4: Fraction of migrants staying / leaving by 2022



Note: Fraction of migrants still in NL by year 2022. All migrants ≥ 18 upon arrival. By year of first move.
Source: CBS Microdata

Table D.6: Income Composition within Origin Groups (%)

	Income			
Origin region	Low (%)	Middle-low (%)	Middle-high (%)	High (%)
Antilles	41.9	18.4	25.6	14.0
China	55.1	16.0	19.0	9.8
EFTA	54.8	10.9	14.5	19.7
EU	50.9	9.3	15.3	24.4
GIPS	46.1	12.7	22.6	18.6
Germany	64.3	10.1	12.2	13.3
India	36.8	10.0	21.9	31.3
Indonesia	72.4	9.7	9.2	8.6
Middle East	66.0	14.0	13.0	7.0
Middle-Eastern Europe excl Poland	47.6	18.3	21.9	12.1
Middle-South America	48.1	19.9	19.1	12.3
Morocco	67.1	11.6	15.8	5.6
Non-EU	53.2	13.0	19.1	14.7
North Africa	57.9	15.4	17.7	9.1
Oceania, Australia, NZ	46.6	9.5	15.0	26.9
Poland	30.5	22.1	39.4	8.0
Remaining Africa	51.1	18.6	21.7	8.6
Suriname	50.0	11.7	23.8	14.6
Turkey	63.3	12.3	16.5	7.9
US/Canada	53.2	7.7	11.9	27.1
remaining Asia	51.0	19.7	18.8	10.4

Notes: This table presents the income composition within regions of origin of immigrants across postcodes and years. The percentages are share of each income tier within the region of origin group.
Sources: CBS Microdata; *Kerncijfers Wijken en Buurten* (KWB).

Table E.7: First stage coefficients with controls

	All migrants		High income		Middle-high income		Middle-low income		Low income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$Z^j_{i,t-1}$	0.531***	0.531***	0.827***	0.828***	0.259**	0.259**	0.613***	0.612***	0.613***	0.615***
	(0.149)	(0.149)	(0.211)	(0.210)	(0.113)	(0.114)	(0.050)	(0.051)	(0.060)	(0.059)
Total origin share (tot_origin_sh)		0.239		-0.049		-0.036		0.102		0.230
		(0.257)		(0.063)		(0.071)		(0.069)		(0.174)
FE: Year	x	x	x	x	x	x	x	x	x	x
FE: PC4	x	x	x	x	x	x	x	x	x	x
Controls	x	x	x	x	x	x	x	x	x	x
N	20,663	20,663	20,663	20,663	20,663	20,663	20,663	20,663	20,663	20,663
Adj. R^2	0.296	0.296	0.368	0.368	0.163	0.163	0.289	0.290	0.168	0.168
IV Wald F statistic	12.754	12.739	15.349	15.537	5.229	5.195	148.231	142.363	105.980	107.257

Note: First stage, as in Equation (3), with a income-type specific SSIV variable, using fixed-1995 shares. One column per income-range of migrant. Outcomes measured in 2006 - 2020. Standard errors in parentheses, clustered by municipality. The regression is weighted by the housing stock. "IV Wald" reports the Wald statistic for weak instruments (Kleibergen-Paap rk statistic) accounting for clustered standard errors.
Statistical significance level: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Source: CBS Microdata

Table E.8: First stage coefficients - Municipality level

	All migrants	High income	Middle-High income	Middle-Low income	Low income
	(1)	(2)	(3)	(4)	(5)
$Z^j_{i,t-1}$	0.924***	1.325***	0.442**	0.833***	0.868***
	(0.142)	(0.328)	(0.175)	(0.030)	(0.075)
FE: Year	x	x	x	x	x
FE: Municipality	x	x	x	x	x
N	4,365	4,365	4,365	4,365	4,365
Adj. R^2	0.309	0.620	0.273	0.565	0.263
IV F statistic	342.4	1318.6	121.5	1943.4	495.2

Note: First stage using municipality-level variables with income-type specific SSIV variable, using 1995 shares. One column per income-range of immigrant. Outcomes measured in 2006 - 2020. Standard errors in parentheses, clustered by municipality. The regression is weighted by the postcode-level population in 2006. "IV F" statistic reports the Cragg-Donald F-statistic under homoskedasticity. "IV Wald" reports the Wald statistic for weak instruments (Kleibergen-Paap rk statistic) accounting for clustered standard errors. Table E.7 shows that the first stage coefficients hardly change when including controls.
Statistical significance level: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Source: CBS Microdata

Table E.9: Effect of a 3-year change of immigrant inflows on 3-year difference in tax assessment value

	Dependent variable: 3-year change tax assessment value									
	All		High income		Middle-High income		Middle-Low income		Low income	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\widehat{\frac{\Delta M^j_{i,t-1}}{Pop_{i,t-2}}}$	-1.107	-14.397*	8.078	-19.120	63.404	-91.060	-10.127	-15.550	3.797*	2.119
	(4.914)	(7.789)	(22.760)	(25.537)	(55.626)	(64.587)	(11.253)	(16.087)	(2.019)	(5.797)
FE: Year	x	x	x	x	x	x	x	x	x	x
FE: PC4	x	x	x	x	x	x	x	x	x	x
Controls:		x		x		x		x		x
N	25,427	17,178	25,427	17,178	25,427	17,178	25,427	17,178	25,427	17,178
Avg outcome	0.0388	0.0388	0.0388	0.0388	0.0388	0.0388	0.0388	0.0388	0.0388	0.0388
Avg regressor	0.0021	0.0021	0.0003	0.0003	0.0004	0.0004	0.0005	0.0005	0.0008	0.0008
SD regressor	0.0064	0.0064	0.0015	0.0015	0.0021	0.0021	0.0022	0.0022	0.0049	0.0049
Impact at mean regressor	-0.0023	-0.0302	0.0024	-0.0057	0.0254	-0.0364	-0.0051	-0.0078	0.0030	0.0017

Note: Endogenous regressor $\frac{\Delta^3 M^j_{i,t-1}}{Pop_{i,t-2}}$: 3-year change of the immigraiton inflow of income group j per neighborhood (4-digit postcode). Outcome: 3-year change of WOZ value per neighborhood (4-digit postcode). Second stage coefficients, as in Equation (1), with an income-type specific SSIV variable, using 1995 shares. The outcome is measured in 2006 - 2020. Standard errors in parentheses, clustered by municipality. Statistical significance level: * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.
Source: CBS Microdata

Table E.10: Excerpt from World Values Survey Wave 6 (2010 - 2014), by size of town

	Size of town (x 1,000)				
	10 - 20	20 - 50	50 - 100	100 - 500	> 500
% who *rather not* have neighbors from the following group:					
Immigrants / foreign workers	19.8	23.7	18.4	17.4	10.1
People of a different race	8	10.4	8	6.2	4.1
People who speak a different language	12.8	18.1	13.3	12.9	7.7
People of a different religion	3.7	3	1.7	2.9	2.4
% respondents who trust					
Neighborhood	74.9	71.7	71.2	64.9	50.3
People of another nationality	33.2	35.3	35.3	38.4	39.7
% respondents to agree for job priority:					
Natives over immigrants	43.9	39.2	35.5	32	24.3
Number of respondents	187	691	4141	419	169

Note: Selection of questions from Wave 6 of the World Values Survey (2010 - 2014).
For the neighborhood questions, survey participants are presented a list of sociodemographic groups and are asked to indicate all groups they would *rather not* have as neighbors. The tables show the percentage of respondents that indicate "immigrants / foreign workers" in their answer. Options are: drug addicts, people of another race, people with AIDS, migrants / foreign workers, homosexuals, people from a different religion, heavy drinkers, unmarried couples living together, people who speak a different language.
For the questions on trust, survey participants indicate their agreement on a 5 point scale from "Trust completely" to "Do not trust at all". The table above shows the percentage of respondents indicating "Trust completely" or "Trust somewhat".
For the question on job security, the table shows the percentage of respondents indicating "Agree" from the options "Agree", "Neither" and "Disagree".
Source: World Values Survey Wave 6: 2010 - 2014